

A DEEP LEARNING-BASED APPROACH FOR INAPPROPRIATE CONTENT DETECTION AND CLASSIFICATION OF YOUTUBE VIDEOS

*A Major Project Report submitted to
JNTU Hyderabad in partial fulfillment
of the requirements for the award of the degree*

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In

ELECTRONICS AND COMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
MALLA REDDY INSTITUTE OF TECHNOLOGY & SCIENCE

(Approved by AICTE New Delhi and Affiliated to JNTUH)

(Accredited by NBA & NAAC with “A” Grade)

An ISO 9001: 2015 Certified Institution

Maisammaguda, Medchal (M), Hyderabad-500100, T. S.

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CERTIFICATE

This is to certify that the Major project entitled **"A Deep Learning – Based Approach for Inappropriate Content Detection and Classification of YouTube Videos"** has been submitted by **RISHAB SAKALE (21S11A0432)**, **BASHETTY HASINI (21S11A0416)**, **BASHETTY AKASH (21S11A0402)** and **KAMMARI ANJANEYULU (21S11A0404)** in partial fulfillment of the requirements for the award of **BACHELOR OF TECHNOLOGY** in **ELECTRONICS AND COMMUNICATION ENGINEERING**. This record of bonafide work carried out by them under my guidance and supervision. **The result embodied in this Major project report has not been submitted to any other University or Institute for the award of any degree.**

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The Major Project work carried out by our team in the Department of Electronics and Communication Engineering, Malla Reddy Institute of Technology and Science, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

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INDEX

CHAPTER	PAGE NO.
ABSTRACT	vi
LIST OF FIGURES	vii
ABBREVIATIONS	viii
1. INTRODUCTION	1
1.1 AIM OF THE PROJECT	2
1.2 SCOPE OF THE PROJECT	2
1.3 OBJECTIVE OF THE PROJECT	2
2. LITERATURE SURVEY	5
2.1 RELATED WORK	10
2.2 MACHINE LEARNING METHOD	10
2.3 DEEP LEARNING METHOD	11
3. EXISITING SYSTEM	14
3.1 DISADVANTAGES OF EXISTING SYSTEM	15
4. PROPOSED SYSTEM	16
4.1 VIDEO PREPROCESSING	16
4.2 DEEP FEATURE EXTRACTION	17
4.3 VIDEO REPRESENTATION AND CLASSIFICATION	18
4.4 Bi-LSTM NETWORK WITH ATTENTION MECHANISM	20
4.5 SOFTMAX CLASSIFIER	22
4.6 OVERALL PROPOSED FRAMEWORK	23
4.7 UML DIAGRAM	23
4.8 USER CASE DIAGRAM	25
4.9 SEQUENCE DIAGRAM	26
5. SYSTEM REQUIREMENTS	27
5.1 EXTERNAL INTERFACE REQUIREMENT	27
5.2 HARDWARE REQUIREMENTS	27
5.3 SOFTWARE REQUIREMENTS	27

6. RESULTS	28
7. ADVANTAGES AND APPLICATIONS	38
7.1 ADVANTAGES	38
7.2 APPLICATIONS	38
8. CONCLUSION & FUTURE SCOPE	39
8.1 CONCLUSION	39
8.2 FUTURE SCOPE	40
9. BIBLIOGRAPHY	41
APPENDIX	43
YUKTI CERTIFICATE	57

ABSTRACT

The exponential growth of videos on YouTube has attracted billions of viewers among which the majority belongs to a young demographic. Malicious uploaders also find this platform as an opportunity to spread up set going visual content, such as using animated cartoon videos to share inappropriate content with children. Therefore, an automatic real-time video content filtering mechanism is highly suggested to be integrated into social media platforms. In this study, a novel deep learning-based architecture is proposed for the detection and classification of inappropriate content in videos. For this, the proposed framework employs an ImageNet pre-trained convolutional neural network (CNN) model known as EfficientNet-B7 to extract video descriptors, which are then fed to bidirectional long short-term memory (BiLSTM) network to learn effective video representations and perform multiclass video classification. An attention mechanism is also integrated after BiLSTM to apply attention probability distribution in the network. These models are evaluated on a manually annotated dataset of 111,156 cartoon clips collected from YouTube videos. Experimental results demonstrated that Efficient Net-BiLSTM (accuracy = 95.66%) performs better than attention mechanism based Efficient Net-BiLSTM (accuracy = 95.30%) framework. Secondly, the traditional machine learning classifiers perform relatively poor than deep learning classifiers. Overall, the architecture of Efficient Net and BiLSTM with 128 hidden units yielded state-of-the-art performance (f1 score = 0.9267). Furthermore, the performance comparison against existing state-of-the-art approaches verified that BiLSTM on top of CNN captures better contextual information of video descriptors in network architecture, and hence achieved better results in child inappropriate video content detection and classification.

LIST OF FIGURES

Fig.No.	Fig. Name	Page No
Fig: 1.1	YouTube cartoon videos depicting different scenarios	4
Fig: 4.1	Proposed framework architecture	19
Fig: 4.2	The architecture of the BiLSTM model	22
Fig: 4.3	User Case Diagram	25
Fig: 4.4	Sequence Diagram	26
Fig: 6.1	Open folders to view training images.	29
Fig: 6.2	Shows violence images for training.	29
Fig: 6.3	Click 'Upload Dataset' to upload.	31
Fig: 6.4	Select and upload 'Dataset' folder.	31
Fig: 6.5	Click 'Dataset Preprocessing' to process images.	32
Fig: 6.6	Dataset: 1047 images. Graph shows labels.	32
Fig: 6.7	EfficientNetB7-BI-LSTM	33
Fig: 6.8	EfficientNetB7-SVM	33
Fig: 6.9	Graph shows EfficientNetB7-BI-LSTM	34
Fig: 6.10	Upload and play video.	35
Fig: 6.11	Detects and classifies as 'Inappropriate.'	35
Fig: 6.12	Classification result shown.	36
Fig: 6.13	Result: 'Safe Content.'	36
Fig: 6.14	Classified as 'Safe'. Test more videos	37

ABBREVIATIONS

AI	:	Artificial Intelligence
BiLSTM	:	Bidirectional Long Short-Term Memory
CNN	:	Convolutional Neural Network
DL	:	Deep Learning
EffNet-B7	:	EfficientNet-B7
LSTM	:	Long Short-Term Memory
ML	:	Machine Learning
NLP	:	Natural Language Processing
RNN	:	Recurrent Neural Network

CHAPTER 1:INTRODUCTION

The creation and consumption of videos on social media platforms have grown drastically over the past few years. Among the social media sites, YouTube predominates as a video sharing platform with plethora of videos from diverse categories. According to YouTube statistics , the global user base of YouTube is over 2 billion registered users and more than 500 hours of video content is uploaded every minute. Consequently, billions of hours of videos are available where users of all age groups can explore generic as well as personalized content.

Considering such a large-scale crowd sourced database, it is extremely challenging to monitor and regulate the uploaded content as per platform guidelines. This creates opportunities for malicious users to indulge in spamming activities by misleading the audiences with falsely advertised content (i.e., video, audio or text). The most disruptive behavior by malicious users is to expose the young audiences to disturbing content, particularly when it is fabricated as safe for them.

Children today spend most of their time on the Internet and the YouTube platform for them has distinctly established itself as an alternative to traditional screen media (e.g., television) [3], [4]. The YouTube press release [5] also confirmed the high popularity of this social media site among younger audiences compared to other age groups, and the reason for this high level of approval is due to fewer restrictions [6]. Unlike television, children can be presented with any type of content on the Internet due to lack of regulations.

Exposing children to disturbing content is considered as one among other internet safety threats (like cyberbullying, cyber predators, hate etc.) [7]. Bushman and Huesmann [8] confirmed that frequent exposure to disturbing video content may have a short-term or long-term impact on children's behavior, emotions and cognition. Many reports [9]–[12] identified the trend of distributing inappropriate content in children's videos.

1.1 AIM OF THE PROJECT

The aim of this project is to develop a deep learning-based system for detecting and classifying inappropriate content in YouTube videos. The system will analyze both video frames and audio transcripts using advanced neural networks, including convolutional neural networks (CNNs) for image analysis and natural language processing (NLP) techniques for text/audio evaluation. The objective is to accurately identify harmful or inappropriate content, such as violence, hate speech, and explicit material, ensuring a safer digital environment. The project will leverage machine learning models trained on large datasets to enhance content moderation and assist platforms in enforcing community guidelines effectively.

1.2 SCOPE OF THE PROJECT

This project aims to develop an AI-based system for detecting and classifying inappropriate content in YouTube videos using deep learning techniques. It involves analyzing both video frames and audio transcripts to identify harmful content, such as violence, hate speech, and explicit material. The system will utilize CNNs for image-based detection and NLP models for text/audio evaluation. Designed for scalability, it can process large datasets and operate in real-time. The project has applications in content moderation, child safety, and corporate compliance. It can be integrated as an API or browser extension to assist platforms in enforcing community guidelines effectively.

1.3 OBJECTIVE OF THE PROJECT

The objective of this project is to develop an AI-powered system that accurately detects and classifies inappropriate content in YouTube videos using deep learning techniques. The system will analyze video frames and audio transcripts to identify harmful content, including violence, hate speech, and explicit material. By leveraging CNNs for image-based detection and NLP models for text/audio analysis, the project aims to enhance automated content moderation. The system will be scalable, capable of real-time processing, and adaptable for integration as an API or browser extension, ensuring a safer digital environment and aiding platforms in enforcing community guidelines effectively.

Hence, this study targets the YouTube multiclass video classification problem by leveraging CNN (EfficientNet-B7) and LSTM to learn video effective representations for detection and classification of inappropriate content. We targeted two types of objectionable content geared towards young viewers, one, which contains violence and the second, which includes sexual nudity connotations. The main contributions of this study are threefold:

- We propose a novel CNN (EfficientNet-B7) and BiLSTM-based deep learning framework for inappropriate video content detection and classification.
- We present a manually annotated ground truth video dataset of 1860 minutes (111,561 seconds) of cartoon videos for young children (under the age of 13). All videos are collected from YouTube using famous cartoon names as search keywords. Each video clip is annotated for either safe or unsafe class. For the unsafe category, fantasy violence and sexual-nudity explicit content are monitored in videos. We also intend to make this dataset publicly available for the research community.
- We evaluate the performance of our proposed CNN-BiLSTM framework. Our multiclass video classifier achieved the validation accuracy of 95.66%. Several other state-of-the-art machine learning and deep learning architectures are also evaluated and compared for the task of inappropriate video content detection.

To summarize, this work can assist any video sharing platform to either remove unsafe video or blur/hide any portion of video involving unsafe content. Secondly, it may also help in the development of parental control solutions on the web via plugins or browser extensions where children inappropriate content filters automatically.

The upcoming sections of the article are outlined as follows: Section II covers the related work in this research area. The methodology of our proposed system is explained in Section III. The experimental setup of the proposed system is presented in Section IV. The results obtained from the experimental setup are

analyzed and discussed in Section V, and finally, Section VI concludes the work and directs some future scope for improvements.

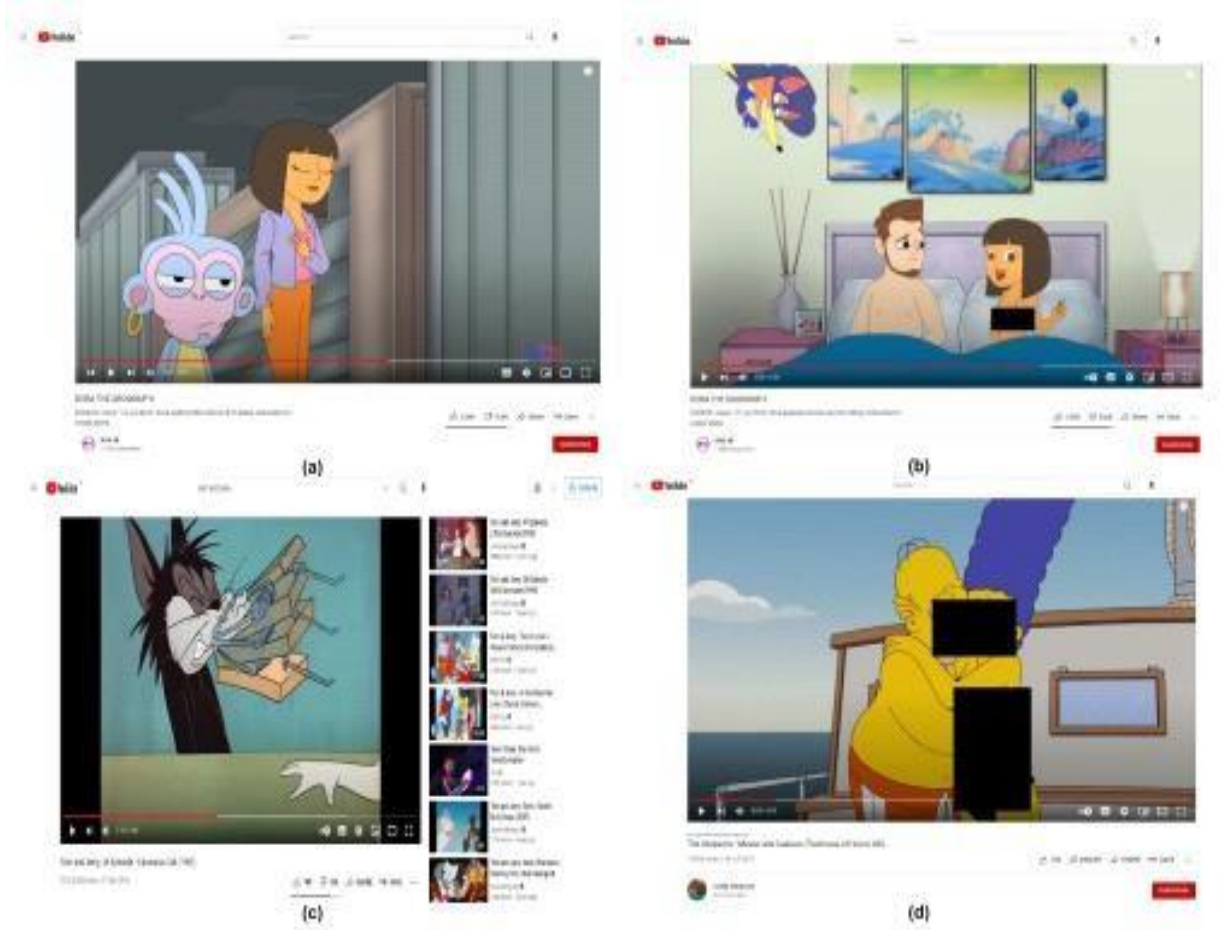


Fig 1.1: Examples of YouTube cartoon videos depicting different scenarios (a) video showing safe content with 8.9 million views, uploaded since 2018 by AOK channel of 1.05 million subscribers, (b) video showing sexual-nudity content with 8.9 million views, uploaded since 2018 by AOK channel of 1.05 million subscribers, (c) video showing fantasy violence content with 32 million views, uploaded since 2016 by WB Kids channel of 19.9 million subscribers, and (d) video showing sexual-nudity content with 14.6k views, uploaded since 2021 by Lovely Simpsons channel of 421 subscribers.

CHAPTER 2: LITERATURE SURVEY

YouTube represents one of the largest scale and most sophisticated industrial recommendation systems in existence. In this paper, we describe the system at a high level and focus on the dramatic performance improvements brought by deep learning. The paper is split according to the classic two- stage information retrieval dichotomy: first, we detail a deep candidate generation model and then describe a separate deep ranking model. We also provide practical lessons and insights derived from designing, iterating and maintaining a massive recommendation system with enormous user-facing impact

YouTube has become a popular digital media platform used by young children. However, concerns have been raised around inappropriate video content and limited quality. A lack of research and theoretical discussion exists on how best to evaluate the quality of YouTube videos made for children. In this study, we reviewed research and developed a set of design principles that informed the production of a YouTube video rubric used to evaluate the quality of YouTube videos targeted at young children 0 to 8 years old.

The four key evaluation criteria used are: Age appropriateness, Content quality, Design features, and Learning objectives. This evaluation tool has the potential to be used by educators to assess the quality of videos for early learning and guide YouTube creators in their production of educational videos for young children. The YouTube video platform has become one of the most popular online applications in the world that young children engage with (Marsh et al. 2019; Rideout 2017), yet compared to other forms of screen media (e.g., television [TV]) this remains rather underexplored (Khan 2017).

There is a paucity of research around video screen media such as YouTube and possible effects on young children’s learning and development (Izci et al. 2019). However, it is clear that YouTube is a form of screen time and through this media, children watch the behaviour of others. This behaviour has the potential to positively or negatively impact on learning, depending on the quality of videos and images presented

YouTube has risen in popularity to become a global phenomenon. Children use YouTube for entertainment and educational purposes. Some benefits of YouTube have been reported such as helping children learn and develop skills and knowledge about their world. However, concerns have been raised about children's exposure to inappropriate and disturbing YouTube content, screen time, and limited video quality.

This article defines what YouTube is, discusses current research on YouTube, and explores what young children are doing with YouTube. It also provides practical strategies and recommendations for parents and teachers about how to use YouTube to support young children's early learning and development at home and school

Objectives To test whether the results of the accumulated studies on media violence and aggressive behavior are consistent with the theories that have evolved to explain the effects. We tested for the existence of both short-term and long-term effects for aggressive behavior. We also tested the theory-driven hypothesis that short-term effects should be greater for adults and long-term effects should be greater for children.

Design Meta-analysis.

Participants Children younger than 18 years and adults.

Main Exposures Violent media, including TV, movies, video games, music, and comic books.

Main Outcome Measures Measures of aggressive behavior, aggressive thoughts, angry feelings, physiological arousal (eg, heart rate, blood pressure), and helping behavior.

The results are consistent with the theory that short-term effects are mostly due to the priming of existing well-encoded scripts, schemas, or beliefs, which adults have had more time to encode. In contrast, long-term effects require the learning (encoding) of scripts, schemas, or beliefs. Children can encode new scripts, schemas, and beliefs via observational learning with less interference and effort than adults.

At the request of UNICEF and its research partners, a team of students at the Human Rights Center at UC Berkeley School of Law spent the Fall 2018 and Spring 2019 semesters researching how artificial intelligence technologies are being used in ways that impact children at home, at school, and at play. We also reviewed and identified the disparate human rights that might be disproportionately impacted, both positively and negatively, by its use. Importantly, while any technology that affects adults will have secondary impacts on children, for the sake of space we focused only on applications that have been designed specifically for children.

As Artificial Intelligence-based technologies become increasingly integrated into modern life, the onus is on companies, governments, researchers, and caregivers to consider the ways in which such technologies impact children's rights. The potential impact of artificial intelligence on children deserves special attention, given children's heightened vulnerabilities and the numerous roles that artificial intelligence will play throughout the lifespan of individuals who are born in the 21st century.

As much of the underlying technology is proprietary to corporations, corporations' willingness and ability to incorporate human rights considerations into the development and use of such technologies will be critical. Governments will also need to work with corporations, parents, children and other stakeholders to create policies that safeguard children's rights and related interests

The exponential growth of videos on YouTube has attracted billions of viewers among which the majority belongs to a young demographic. Malicious uploaders also find this platform as an opportunity to spread upsetting visual content, such as using animated cartoon videos to share inappropriate content with children. Therefore, an automatic real-time video content filtering mechanism is highly suggested to be integrated into social media platforms. In this study, a

novel deep learning-based architecture is proposed for the detection and classification of inappropriate content in videos.

Secondly, the traditional machine learning classifiers perform relatively poor than deep learning classifiers. Overall, the architecture of EfficientNet and BiLSTM with 128 hidden units yielded state-of-the-art performance (f1 score = 0.9267). Furthermore, the performance comparison against existing state-of-the-art approaches verified that BiLSTM on top of CNN captures better contextual information of video descriptors in network architecture, and hence achieved better results in child inappropriate video content detection and classification.

The volatile growth of social media content on the Internet is revolutionizing content distribution and social interaction. Social media exploded as a category of online discourse where people create content, share it, bookmark it and network it at prodigious rate. Examples comprise Facebook, MySpace, Youtube, Instagram, Digg, Twitter, Snapchat and others. Since it is easy to reach, use and the high velocity of spreading information among users.

The internet as it is at present is made up of a vast array of protocols and networks where traffickers can anonymously share large volumes of illegal material amongst each other from locations with relaxed or non-existent laws that prohibit the possession or trafficking of illegal material. In this paper, a review of applications of deep networks techniques has been presented.

Hence, the existing literature suggests that we do not lose sight of the current and future potential of applications of deep network techniques. Thus, there is a high potential for the use of Convolutional Neural Networks (CNN) for violence video classification, which has not been fully investigated and would be one of the interesting directions for future research in video classification.

Due to tech-savvy offenders and technological advances, continuous efforts in keeping up with current developments are crucial and need to be considered in the implementation of detection algorithms. The present research provides a comprehensive synthesis and an interpretation of the current research accomplishments and challenges in the CSAM detection domain, explicitly considering the dimensions of policy and legal framework, distribution channels, and detection applications and implementations.

Among other aspects, it reveals and aggregates knowledge related to image hash database, keywords, web-crawler, detection based on filenames and metadata, and visual detection. The findings suggest that CSAM detection applications yield the best results if multiple approaches are used in combination, such as deep-learning algorithms with multi-modal image or video descriptors merged together. Deep-learning techniques were shown to outperform other detection methods for unknown CSAM.

This paper analyzes emerging shifts in YouTube, advertising, and children's digital media industries through a case study of Pocket Watch, a digital-first production and distribution studio built exclusively for YouTube child stars. Our analysis reveals the company's strategic use of legacy media industry power, networks, and expertise to transform YouTube stars into global brands through the creation of toy, clothing, and lifestyle product lines across several industries.

We further argue that Pocket Watch's newly formed advertising division, Clock Work, exploits its child partners through problematic native advertising and host selling practices. The strategies implemented by Pocket Watch and other similar emerging companies may therefore act as a litmus test for how governmental regulation and platform policy changes will impact the evolving landscape of children's digital media as commercial forces increasingly groom a growing number of young children to shift from YouTube stars to global brands.

2.1 RELATED WORK

The explosion of multimedia data on YouTube has presented a lot of opportunities for researchers [33]. However, the challenging task is to find an optimal technique to understand the context of videos. In both computer vision and machine learning, video classification is studied extensively as one of the fundamental approaches for video understanding [34]. The problem of detecting inappropriate video falls in the category of video classification or event detection problem.

2.2 MACHINE LEARNING METHOD

Most of the earlier studies used hand-crafted features on images or video-level data in identifying the discriminative patterns of inappropriate content. The skin (i.e., skin color) [35] and motion information based features were used for nudity or pornography detection [26], [36].

The multimodal approach was also followed by fusing different modalities of data (i.e., audio, video) with skin and motion-based features. Rea et al. [37] proposed a periodicity-based audio feature extraction method which was later combined with visual features for illicit content detection in videos. The machine learning algorithms are usually employed as classifiers Liu et al. [38] classified the periodicity-based audio and visual segmentation features through support vector machine (SVM) algorithm with Gaussian radial basis function (RBF) kernel.

Tang et al. [42] proposed a pornography detection system—PornProbe, based on a hierarchical latent Dirichlet allocation (LDA) and SVM algorithm. This system combined an unsupervised clustering in LDA and supervised learning in SVM, and achieved high efficiency than a single SVM classifier. Lee et al. [43] presented a multilevel hierarchical framework by taking the multiple features of different temporal domains.

Lopes et al. [44] worked with the bag-of-visual features (BoVF) for obscenity detection. Kaushal et al. [21] performed supervised learning to identify the child unsafe content and content uploaders by feeding the machine learning classifiers (i.e., random forest, K-nearest neighbor, and decision tree) with video-level, user-

level and comment-level metadata of YouTube Reddy et al. [45] handled the explicit content problem of videos through text classification of YouTube comments. They applied bigram collocation and fed the features to the naïve Bayes classifier for final classification.

2.3 DEEP LEARNING METHOD

In contrast to machine learning algorithms, there is a growing trend of using deep learning architectures to learn the video-based feature representations in video classification. The study of Ngiam et al. [46] is the first deep learning-based research that processed different data modalities such as video, image, audio and text, and performed greedy layer-wise training of restricted Boltzmann machine (RBM) model.

Karpathy et al. [29] showed multiclass video classification results on a large-scale video dataset (ImageNet) by using the convolutional neural network. This study is followed by the research of Yue-Hei Ng et al. [32] in which information across the full-length duration of videos is examined. The LSTM model is employed on top of frame-level CNN activations for better video classification.

Wehrmann et al. [48] reported the best accuracy results with CNN-LSTM than other baselines on the NPDI pornography dataset [24]. Perez et al. [49] classified pornography videos by deploying CNNs with static and motion information (i.e., MPEG motion vectors).

This study yielded the same accuracy scores when static and motion information are combined by late fusion or mid-level fusion (96.4%), but the performance degraded in early fusion (90.5%) Aldahoul et al. [50] explored and compared different state-of-the-art deep learning approaches and found that the pretrained CNN model (EfficientNet-B7) based features with SVM classifier performed better in unsafe video detection.

In the literature, many studies explored different YouTube data modalities (i.e., text, audio, image, and video) individually for inappropriate content detection. Yenala et al. [51] proposed a novel CNN- BiLSTM model for automatic detection and filtering of the inappropriate text in query suggestions of YouTube search.

Trana et al. [52] investigated naïve Bayes, SVM and CNN classifiers for harassment detection in YouTube comments Dadvar and Eckert [53] identified cyber bullying in YouTube comments by experimenting with four deep learning models (i.e., CNN, LSTM, BiLSTM and BiLSTM with attention) using GloVe, SSWE and random word embedding's.

In this study, the BiLSTM with an attention mechanism performed better than conventional machine learning algorithms. Mohaouchane et al. [54] performed an automatic detection of YouTube offensive comments using the CNN-BiLSTM model. Alshamrani [55] used an ensemble classification model to detect the age-inappropriate comments posted on YouTube videos. Later on, they extended the work

Features from three different sources (video metadata, audio transcripts and thumbnail) are extracted and fed to individual classifiers such as metadata to SVM with linear kernel, thumbnails to the random forest and audio transcripts to RNN-based gated recurrent unit (GRU) classifier Hou et al. [10] recognized the bloody videos by combining the audio-visual features and passed them to the CNN- LSTM model.

Ali and Senan [11] also explored the audio-visual features of YouTube videos with the DNN model for violence classification. Sumon et al. [58] proposed the CNN-LSTM model for identifying the violent crowd flow in YouTube videos Alghowinem [59] showed a multimodal approach for inappropriate content filtering of the YouTube Kids application.

Papadamou et al. [20] studied an Elsagate phenomenon in the YouTube Kids application by using YouTube video metadata such as video title, tags, statistics (likes/dislikes, view count, comments count), style features (same as proposed in [21]) and thumbnails. The authors deployed a fully connected neural network model for statistics and style features, CNN for thumbnail features, and LSTMs for video title and tags features. Vitorino et al. [60] explored transfer learning and fine-tuning with CNN for sexually exploitative imagery of children (SEIC) detection.

Ishikawa et al. [61] proposed an end-to-end pipeline for combating the Elsagate content in YouTube videos. They evaluated the classification performances of mobile-platform based different deep learning architectures such as GoogLeNet, SqueezeNet, NASNet, and MobileNetV2. Tahir et al. [22] processed multimodal data with the CNN-LSTM framework for disturbing and fake embedded content in videos. Lastly, Singh et al. [62] performed a fine-grained approach for child unsafe video content detection.

They employed the LSTM autoencoder to learn video representations from CNN (VGG-16) feature descriptors. The abundance of literature is available for inappropriate video content detection by using hand-crafted or deep learning feature extraction techniques. These studies have performed binary classification to classify the videos as safe or unsafe.

However, it lacks the research of detecting or classifying different categories of disturbing content in real-time YouTube videos, particularly in children-oriented videos. Secondly, the existing studies have not explored the BiLSTM network for inappropriate video content detection. This paper addresses the aforementioned problems by working with EfficientNet-B7 and BiLSTM-based deep learning framework for children inappropriate video content detection and classification.

CHAPTER 3:EXISTING SYSTEM

In the current scenario, content moderation on platforms like YouTube relies on a combination of manual review, rule-based filtering, and AI-driven techniques to detect inappropriate content. However, these existing methods have limitations in terms of accuracy, efficiency, and scalability.

One of the primary approaches used in the existing system is manual content moderation, where human reviewers analyze reported videos to determine if they violate platform policies. YouTube has a team of content moderators who review flagged videos based on community guidelines. However, this process is time-consuming, labor-intensive, and prone to human bias. Additionally, due to the massive volume of videos uploaded every minute, manual review cannot keep up with the growing content, leading to delayed actions on harmful videos.

Another commonly used approach is rule-based filtering, where platforms use predefined keyword lists, metadata analysis, and pattern recognition techniques to detect inappropriate content. This method works by scanning video titles, descriptions, comments, and transcripts for specific words or phrases associated with harmful content, such as hate speech, profanity, or explicit language. While this approach can identify obvious violations, it is ineffective against more sophisticated content manipulation techniques, such as disguised hate speech, misleading titles, or visually inappropriate content that bypasses text-based filters.

One significant drawback of the existing system is its limited multimodal analysis, where most detection methods focus on either text-based filtering or individual image frames, rather than considering video, audio, and text together. Deep learning techniques that integrate multiple modalities (video frames, speech, and metadata) are still evolving and are not fully implemented in mainstream content moderation systems.

The current content moderation system on YouTube relies on a combination of manual review, rule-based filtering, and AI-based techniques. However, it faces significant challenges, including high labor costs, scalability issues, lack of contextual understanding, and inefficiencies in real-time content detection. To address these limitations, a deep learning-based approach that integrates advanced neural networks, real-time analysis, and multimodal data processing is needed to enhance the accuracy and efficiency of inappropriate content detection on YouTube.

3.1 Disadvantages of Existing System:

- ❑ Slow and Inefficient Manual Review
- ❑ Limited Accuracy of Rule-Based Filtering
- ❑ Challenges in AI-Based Detection
- ❑ Limited Multimodal Analysis
- ❑ Scalability and Real-Time Limitations
- ❑ Inconsistencies and Bias in Decision-Making
- ❑ Lack of Adaptive Learning for Evolving Content

CHAPTER 4: PROPOSED SYSTEM

The proposed methodology provides a system to resolve the problem of disturbing content in videos. This work employs deep learning architecture which has already been applied successfully in several applications for video classification problems. As shown in Fig. 2, the proposed system is divided into three main modules, namely

- (1) video pre-processing,
- (2) deep feature extraction, and
- (3) video representation and classification.

In the video pre-processing stage, the collected YouTube videos are pre-processed to remove all irrelevant or missing video information. It also rescales the extracted frames of each video clip into fixed dimensions (224×224). The pre-processed video frames of each video clip are forwarded as an input to an ImageNet pre-trained EfficientNetB7 model for feature extraction.

The extracted features are experimented with the BiLSTM network to learn effective video representations, which subsequently passed to the fully connected and softmax layers for final video classification. Furthermore, the detailed descriptions of each step are explained in the following subsections.

4.1 VIDEO PREPROCESSING

Video preprocessing plays an important role in deep learning techniques, as it helps in acquiring the relevant features for better video classification. In this work, a video (V_i) is first represented as N small segments of videos referred to as video *clips* ($ci1, ci2; : : : ; ciN$) with one-second length each.

These video clips are labeled through a manual annotation process where all clips with incomplete information or video context are ignored. After splitting and labeling of video clips, it is noticed that the initial 3-4 frames of each video clip have a piece of information from the previous clip of the same video. Therefore, considering an average video frame rate of 23-24 frames per second (fps), each of the j th video clip (ci_j) is sampled at the frame rate of 22 fps by ignoring some initial frames.

The last frame is duplicated for all video clips containing fewer frames than an average frame rate of a video. Overall, the frames are represented as $f_{1i;j}, f_{2i;j}, \dots, f_{22i;j}$, where $f_{ki;j}$ means k th frame of video clip c_{ij} . Finally, the selected frames of each video clip are rescaled to a fixed resolution of 224×224 pixels that correspond to the input size of the pre-trained convolutional neural network model

4.2 DEEP FEATURE EXTRACTION

In this module, the deep features of preprocessed video frames are extracted by using a deep learning model with advanced architecture. Instead of training an entire CNN model from scratch, this study employed a pre-trained CNN architecture known as Efficient Net for the extraction of visual representations from video frames.

4.2.1 EFFICIENT-NET

The Efficient Net model is a convolutional neural network model and scaling method that uniformly scales network depth, width and resolution through compound coefficient. It is trained on a large-scale ImageNet dataset with 1.3 million images from 1000 object classes [63]. Tan and Le [64] reported state-of-the-art accuracy of EfficientNet on ImageNet dataset with much smaller and faster inference than best existing CNN models.

The baseline network of EfficientNet is referred to as B0, whereas other scaling networks include B1, B2, B3, B4, B5, B6 and B7 respectively. In general, all scaling networks show better accuracy but with the cost of FLOPS. The proposed framework included EfficientNet-B7 by working with preprocessed extracted frames ($f_{i,j,1}, f_{i,j,2}, \dots, f_{i,j,22}$) of each video clip c_{ij} as an input.

The EfficientNet module performed feature extraction with the transfer learning technique in which each input frame with dimensions of $224 \times 224 \times 3$ (image_width x image_height x RGB_channel) is processed through a stack of 813 layers.

4.3 VIDEO REPRESENTATION AND CLASSIFICATION

The third stage of pipeline trained bidirectional LSTM network with supervised learning so that effective video representations can be learnt from feature descriptors of video clips. Subsequently, the proposed system is added with two fully connected layers for acquiring the final video classification results.

4.3.1 BiLSTM NETWORK

The recurrent neural networks produce good network performance in modeling the hidden sequential patterns of time-series data. However, the vanishing gradient problem hampers an update of network parameters during the back propagation process. It is usually resolved by using two variations of RNNs, which are: LSTM and gated recurrent unit (GRU).

Conceptually, the network structure of LSTM is as same as RNN, but a special unit “memory cell” is introduced in LSTM to replace the update process of RNN. The memory cell of LSTM maintains information for a longer duration of time. Considering the current input vector x_t , the last hidden state h_{t-1} , and the last memory cell state c_{t-1} , the following equations are used to implement an LSTM Smodel:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (2)$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \quad (4)$$

$$h_t = o_t \cdot \tanh(c_t) \quad (5)$$

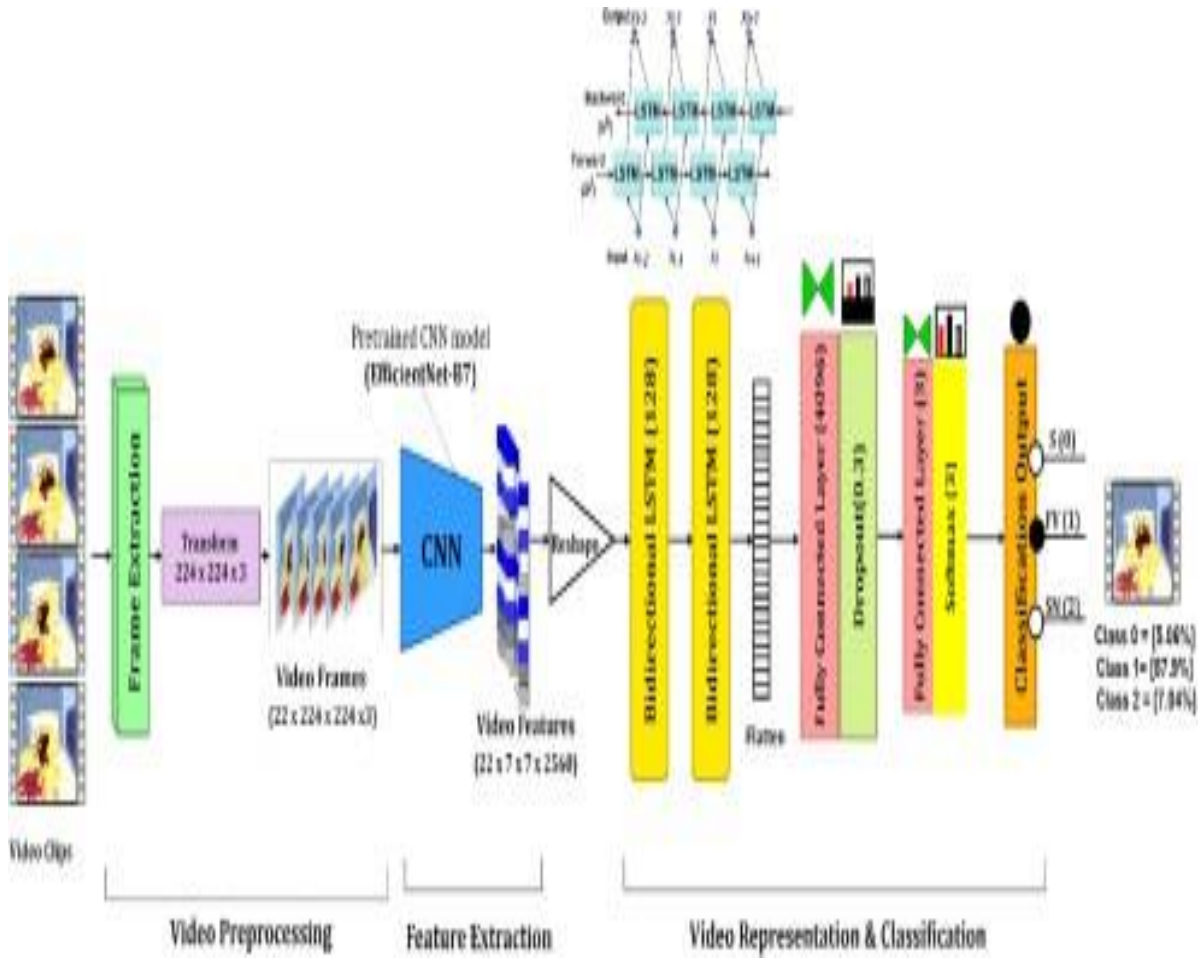


Fig 4.1. The proposed framework architecture works into three stages for children inappropriate video content detection and classification. In the first stage, video clips are preprocessed to discard irrelevant video frames and transform the selected frames into fixed-size images (224,224,3). Next, the frames are fed to EfficientNet-B7 to get feature vectors. All feature vectors are reshaped and passed into the two-layer stack of BiLSTMs for video representation. A fully connected layer followed by an output layer with softmax activation is integrated to return the probabilities of each video clip against the three classes including safe (class 0), fantasy violence (class 1) and sexual-nudity (class 2).

where, σ represents the sigmoid activation function; i , f , c , and o denote input gate, forget gate, memory cell state and output gate at time t , respectively. W and b denote the weights and bias vector. Considering the video classification problem, one potential drawback of LSTM is that it captures the past context only.

For getting the full context of any video, it is important to consider both directions i.e., past and future context of the video. Therefore, the bidirectional LSTM appears to be a suitable option in video classification as it preserves the information in both directions, as shown in Fig. 3. In BiLSTM, there are two distinct hidden layers referred to as forward hidden layer (h^f) and backward hidden layer (h^b).

The forward hidden layer h^f considers input vector x_t in ascending order i.e., $t = 1, 2, 3, \dots, T$, and backward hidden layer h^b in descending order i.e., $t = T, T-1, T-2, \dots, 1$. Lastly, the output y_t is generated by combining the results of h^f and h^b . Following equations are used to implement the BiLSTM model:

$$h_t^f = \tanh(W_{xh}^f x_t + W_{hh}^f h_{t-1}^f + b_h^f) \quad (6)$$

$$h_t^b = \tanh(W_{xh}^b x_t + W_{hh}^b h_{t+1}^b + b_h^b) \quad (7)$$

$$y_t = W_{hy}^f h_t^f + W_{hy}^b h_t^b + b_y \quad (8)$$

4.4 BiLSTM NETWORK WITH AN ATTENTION MECHANISM

A neural network architecture with an attention mechanism decides when to look into data (or in this case, segments of videos) by automatically giving a high level of focus to feature vectors with most valuable information than the feature vectors with less valuable information. The architecture of BiLSTM with an attention mechanism is depicted in Fig. 4. Considering the final hidden state of i -th BiLSTM as h_{it} , which is computed as:

$$h_{it} = [h_t^f, h_t^b] \quad (9)$$

Then, the attention mechanism is computed by using the following equations

$$e_{it} = \tanh(W_a h_{it} + b_a) \quad (10)$$

$$a_{it} = \frac{\exp(e_{it})}{\sum_{j=1}^T \exp(e_{jt})} \quad (11)$$

$$v_t = \sum_{i=1}^T a_{it} . h_{it} \quad (12)$$

The attention mechanism assigns attention weight a_{it} to the i -th BiLSTM output vector at time t , as calculated in equation 11. W_a and b_a represents the weight and bias from the attention layer.

Finally, the output from attention layers generates an attention vector v_t , which is calculated as a weighted sum of the multiplication between attention weight a_{it} and i -th BiLSTM output vector at time t .

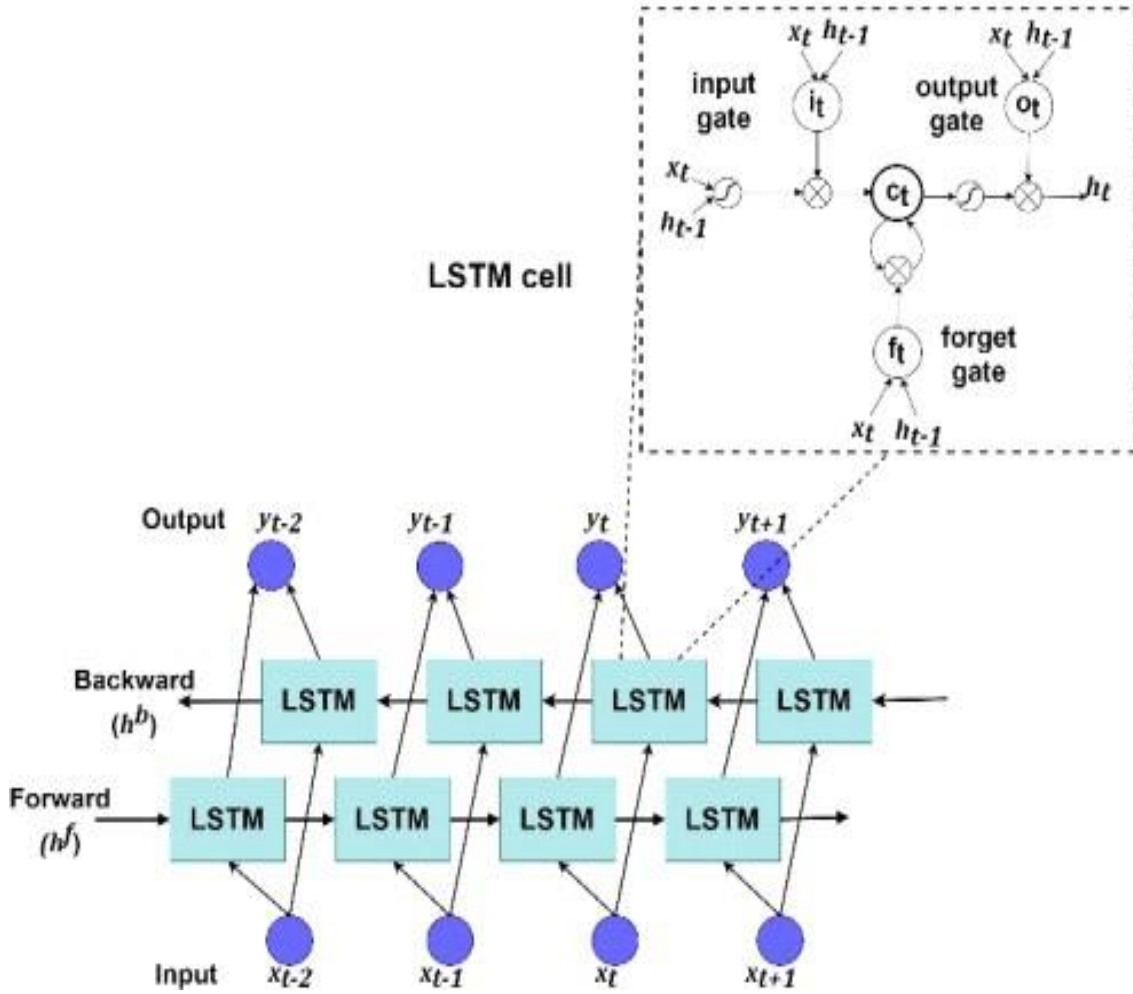


Fig 4.2. The architecture of the BiLSTM model

4.5 SOFTMAX CLASSIFIER

The softmax activation function in an output layer of any deep learning model is considered as a softmax classifier.

To classify each video clip into one of three classes (i.e., fantasy violence, sexual-nudity and safe), the proposed model integrated a softmax activation function in the last fully connected layer to determine the relative probability of three output units. The softmax activation function (σ) is calculated as:

$$\sigma(z_i) = \frac{\exp(z_i)}{\sum_{c=0}^{N-1} \exp(z_c)} \quad (13)$$

4.6 OVERALL PROPOSED FRAMEWORK

The general architecture of the proposed framework for multiclass video classification is illustrated in Fig. 2. The input is a sequence of 22 frames of fixed resolution with $224 \times 224 \times 3$ pixels. These frames are processed through EfficientNet-B7 for extracting features and generating the feature vector of $22 \times 7 \times 7 \times 2560$ shape.

The high-level features are reshaped and directed towards the two-layer stack of BiLSTM network. The flattening layer is added to transform the feature representations into a 1-dimensional vector. Subsequently, a fully connected layer (or dense layer) of 4096 neurons with rectified linear unit (ReLU) activation function is added.

As a fully connected layer generates a wide range of probabilities by connecting all inputs of one layer to every activation unit of the next layer; therefore, a dropout of 0.3 is carried out to prevent this model from an overfitting problem. Finally, the softmax output layer with 3 neurons gives the final classification scores. The algorithmic steps for training and validation of the proposed framework are presented in Algorithm 1.

4.7 UML DIAGRAMS:

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Integrate best practices.

4.8 USER CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

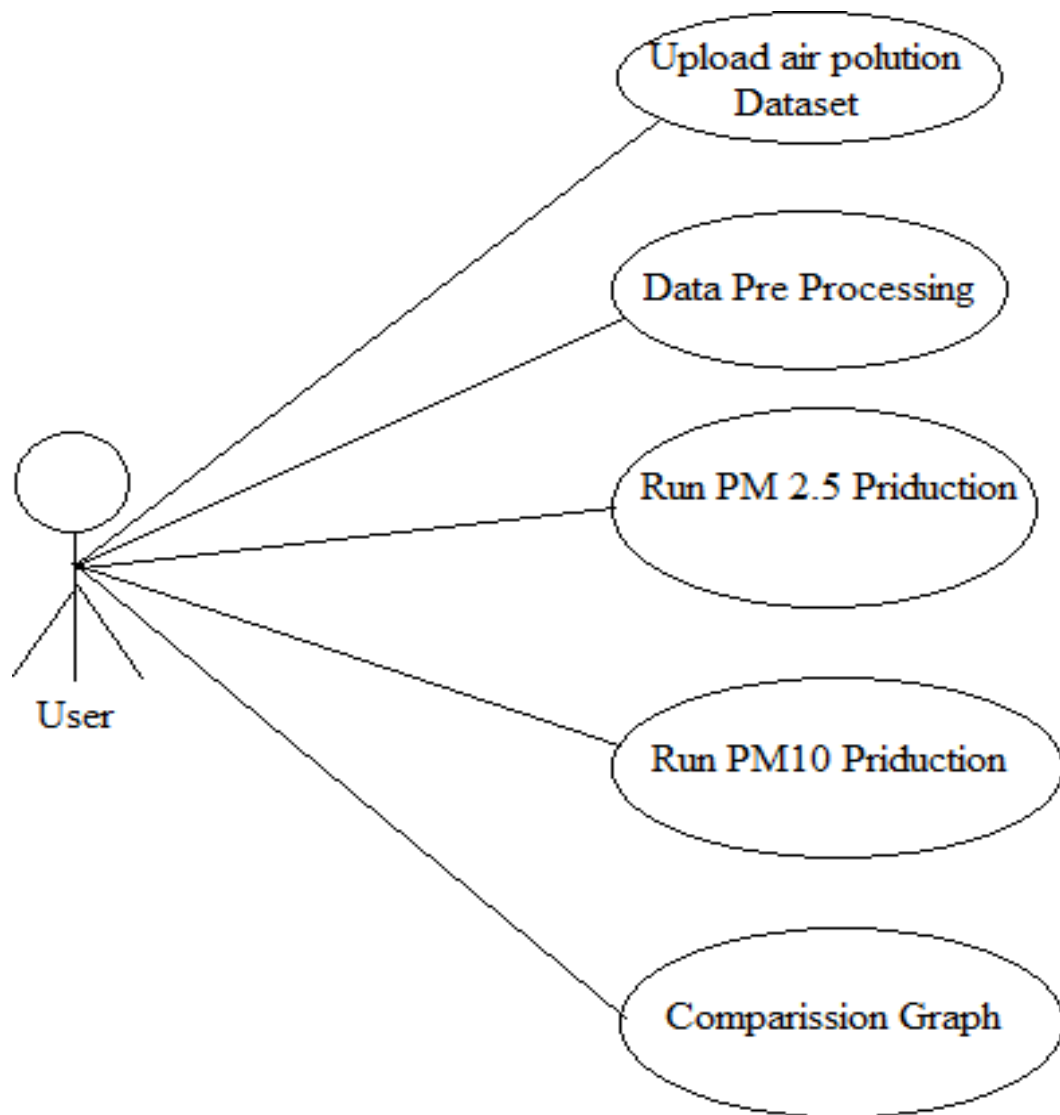


Fig : 4.8 USER CASE DIAGRAM:

4.9 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams

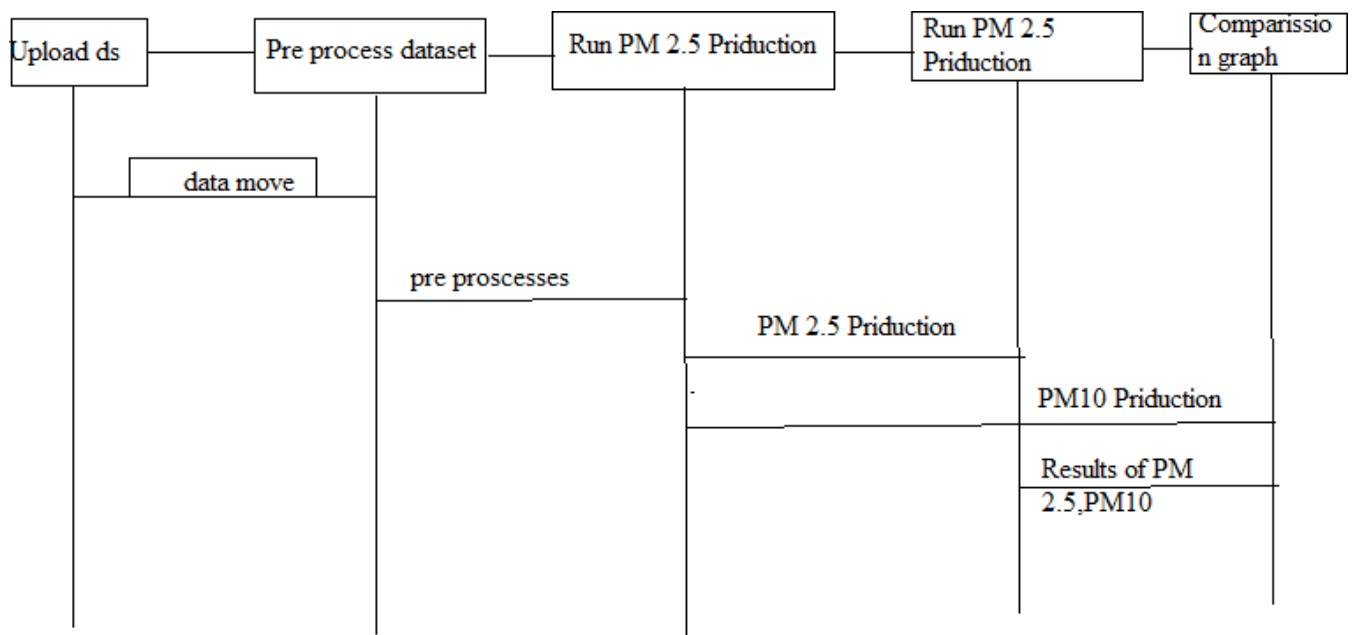


Fig: 4.9 SEQUENCE DIAGRAM

CHAPTER 5: SYSTEM REQUIREMENTS:

5.1 EXTERNAL INTERFACE REQUIREMENTS:

User Interface

This program's touch screen is a browser Python Graphic User Experience.

Hardware Interfaces

Pycharacteristics are used to allow the users to communicate with the console.

Software Interfaces

Python is the needed software.

Operating Environment

Windows XP is the operating system

5.2 HARDWARE REQUIREMENTS:

Processor	Pentium –IV
Speed	1.1 Ghz
RAM	512 MB(min)
Hard Disk	20 GB
Key Board	Standard Windows Keyboard
Mouse	Two or Three Button Mouse
Monitor	SVGA

5.3 SOFTWARE REQUIREMENTS:

Operating System	Windows 10/11
Programming Language	Python (python 3.7.0)

CHAPTER 6: RESULTS AND DISCUSSION

Now-a-days YouTube content are access by all age group of peoples as this provide digital entertainment content on various topics such as Sports, Religious, Movies and many more. This channel provides vast amount cartoon entertainment for KIDS. Some malicious users are taking advantage of this digital content to spread inappropriate content for kids the in the form of cartoons. Such inappropriate content may put bad influence on growing kids and need a technique to prevent such content before showing to kids.

To prevent such content many machine learning and deep learning algorithms are introduced but their inappropriate content detection and classification accuracy is not up to the mark. To overcome from this issue author of this paper employing combination of CNN and BI-LSTM algorithm to detect inappropriate content.

Pre-trained EfficientNetB7 CNN algorithm is employed to extract features from the YouTube video images and then retrained with BI-LSTM algorithm to enhance prediction accuracy.

In propose work a snovel deep learning-based architecture is proposed for the detection and classification of inappropriate content in videos. For this, the proposed framework employs an ImageNet pre-trained convolutional neural network (CNN) model known as EfficientNet-B7 to extract video descriptors, which are then fed to bidirectional long short-term memory (BiLSTM) network to learn effective video representations and perform multiclass video classification. An attention mechanism is also integrated after BiLSTM to apply attention probability distribution in the network.

Two algorithms are trained on YouTube video annotated images such as Attention based BI- LSTM and EfficientNetB7 based BI-LSTM and in both algorithms EfficientNetB7 is giving better accuracy. On same dataset we have experiment with existing SVM algorithm but its accuracy is less than propose EfficientNetB7-BILSTM algorithm.

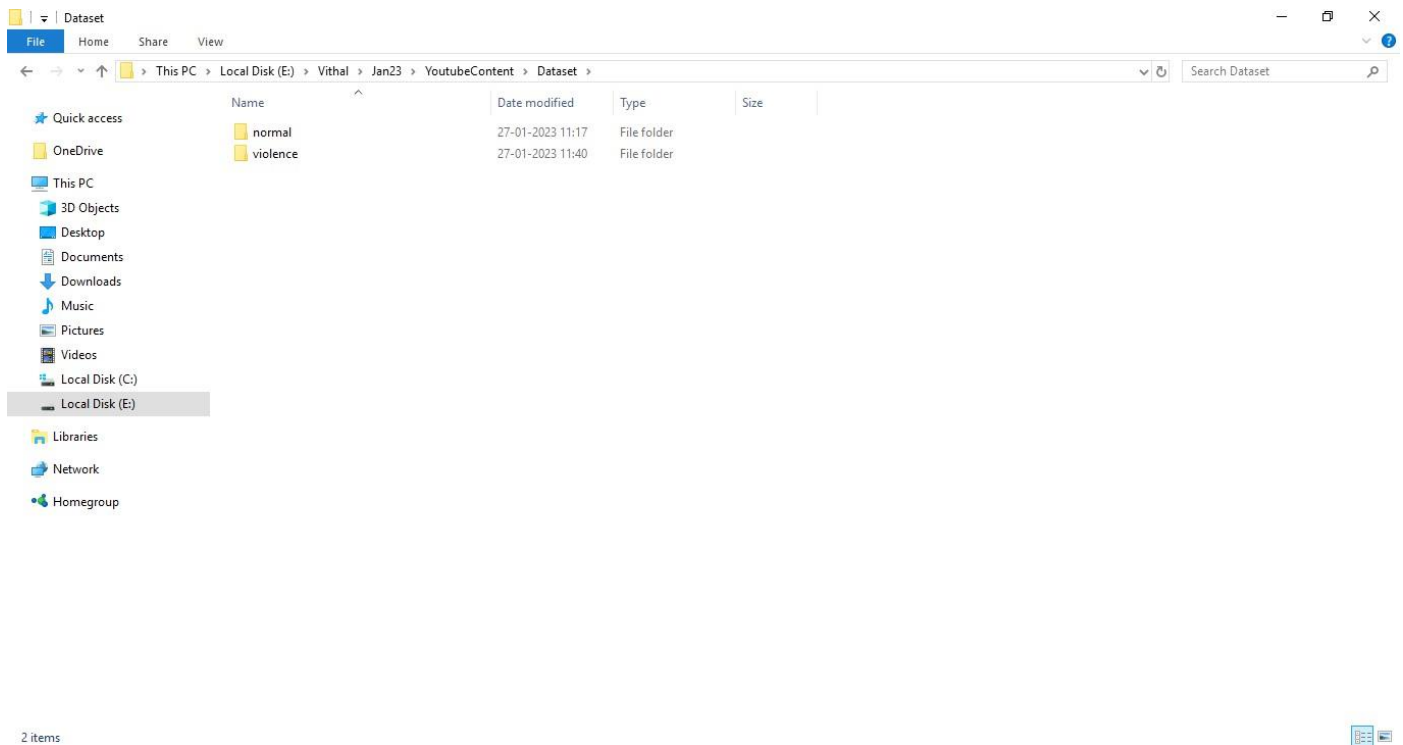


Fig : 6.1 - In above screen we have two folders and juts go inside any folder to view training images

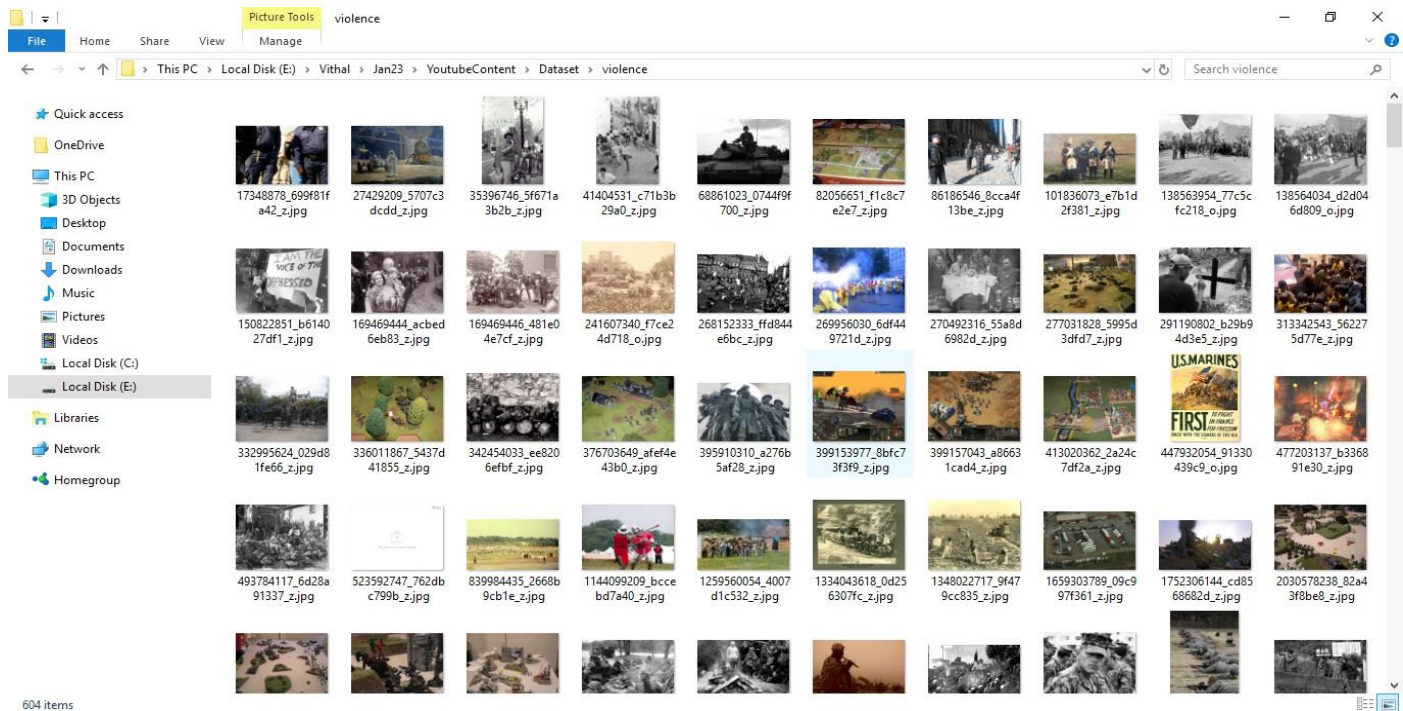


Fig : 6.2 - In above screen we can see violence images used to train algorithms

To implement this project we have designed following modules

- Upload YouTube Normal & Inappropriate Content Dataset: using this module we will upload YouTube dataset images to application
- Dataset Preprocessing: using this module we will read all images and then resize all images to equal size and then normalize image pixel values and then shuffle the dataset
- Run Propose DL-BILSTM-GRU Algorithm: using this module we will split dataset into train and test and then input 80% training data to Pre-Trained CNN (EfficientNetB7) algorithm to extract digital content from images and then those features will get retrained with BI-LSTM algorithm to train a model. Trained model will be applied on 20% test data to calculate prediction accuracy
- Run EfficientNet-SVM Algorithm: EfficientNetB7 features will get retrained with existing SVM algorithm and then calculate prediction accuracy
- Comparison Graph: using this module we will plot accuracy comparison graph between psopose EfficientNetB7-BILSTM and EfficientNetB7-SVM.
- Inappropriate Content Prediction from Test Video: using this module we will upload any YouTube and if video contains fighting or violence then application will predict as 'Inappropriate Content' otherwise will predict SAFE content.

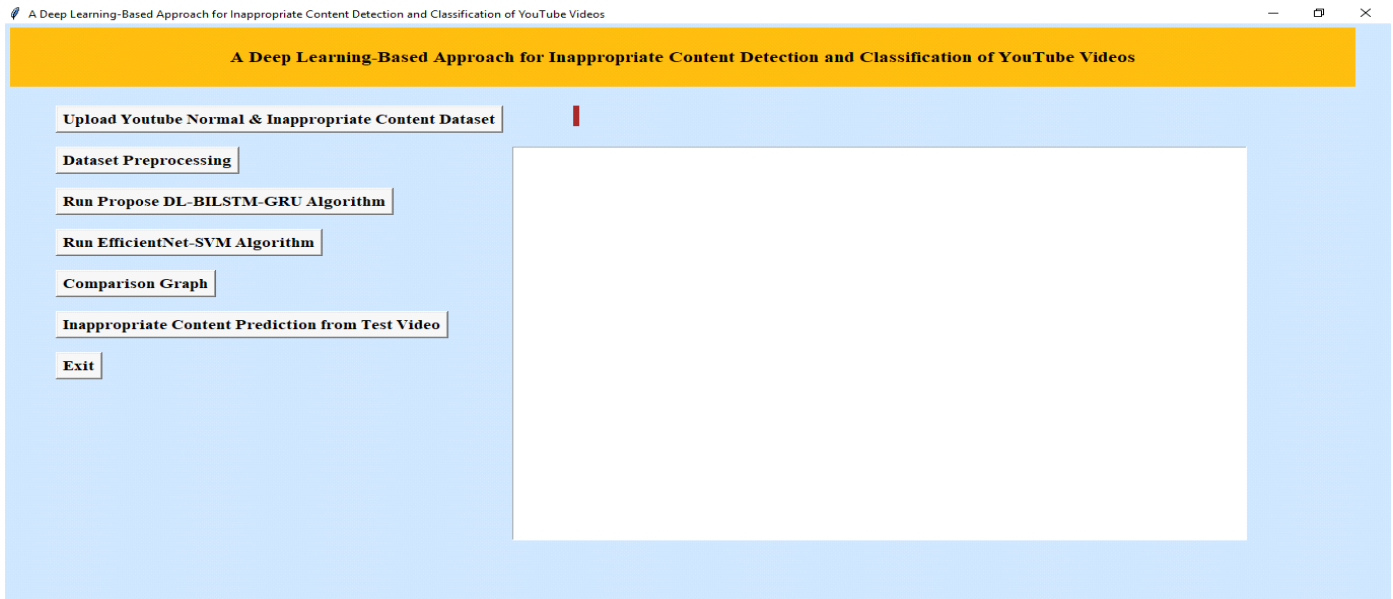


Fig : 6.3 - In above screen click on ‘Upload YouTube Normal & Inappropriate Content Dataset’ button to upload dataset and get below output

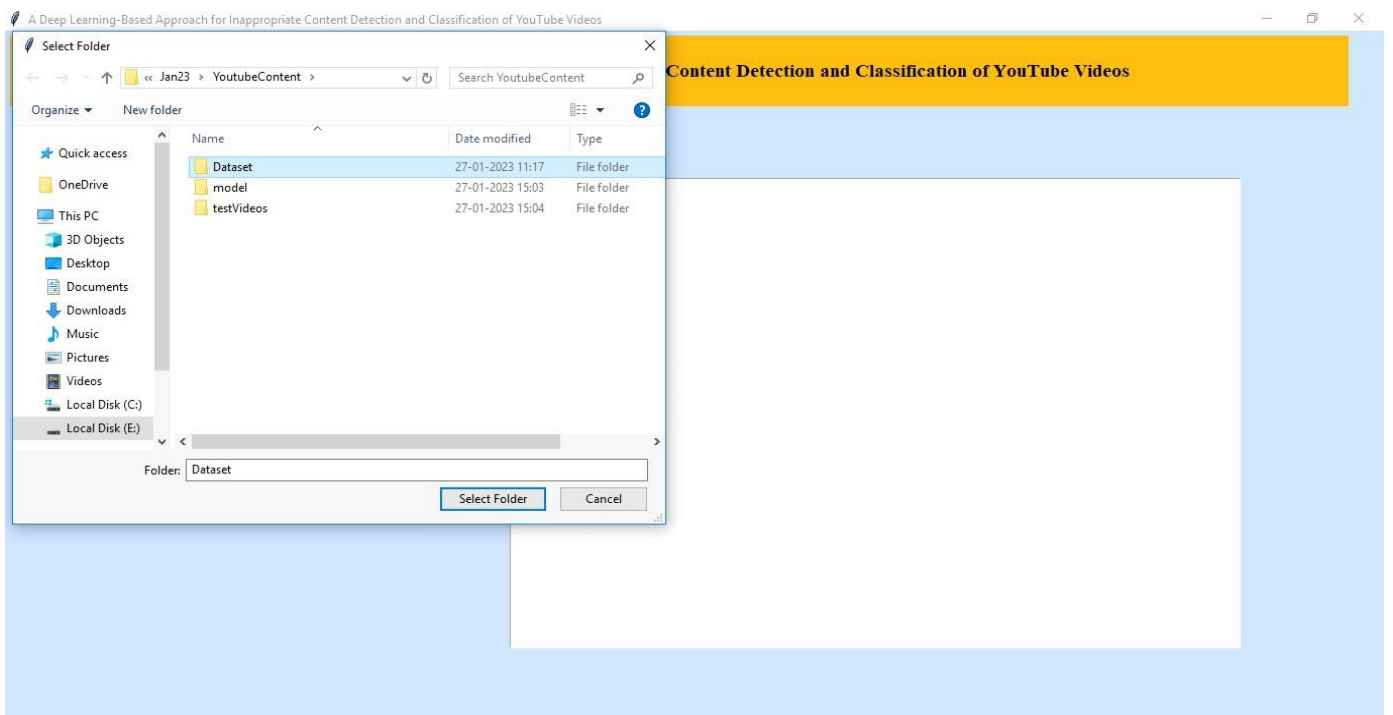


Fig : 6.4 - In above screen selecting and uploading entire ‘Dataset’ folder and then click on ‘Select Folder’ button to load dataset and get below output

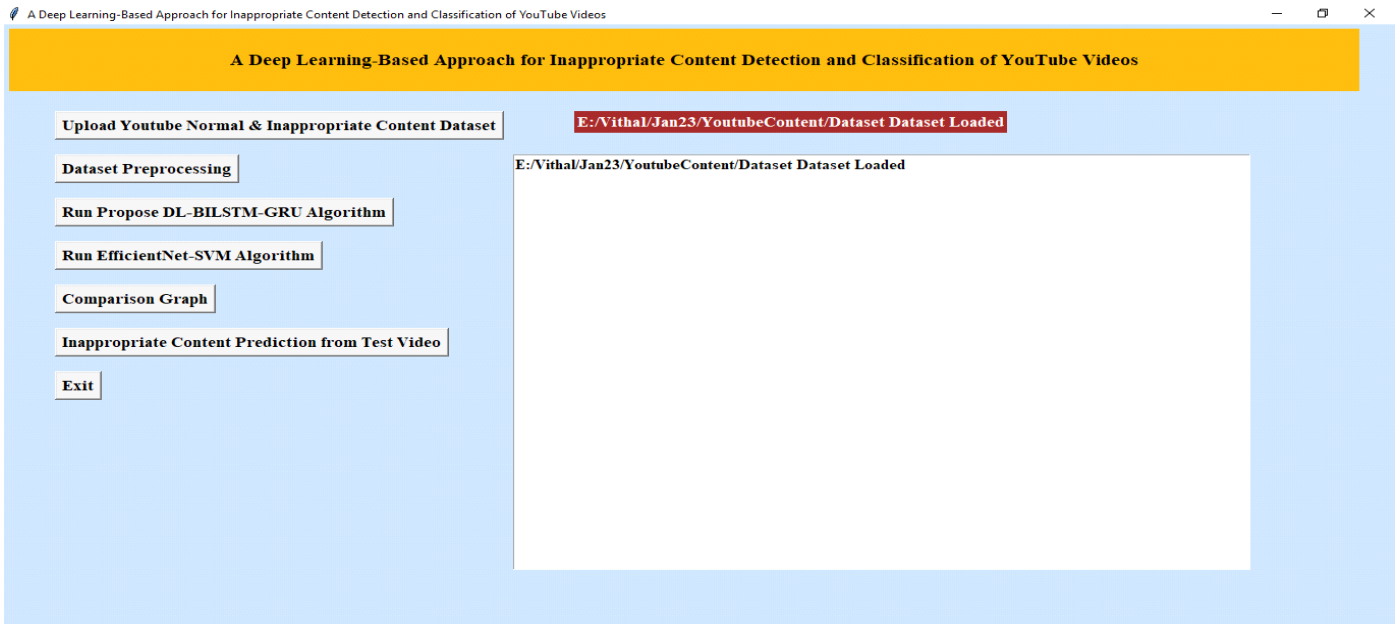


Fig : 6.5 - In above screen dataset loaded and now click on ‘Dataset Preprocessing’ button to read all images and then processes those images for training and get below output.

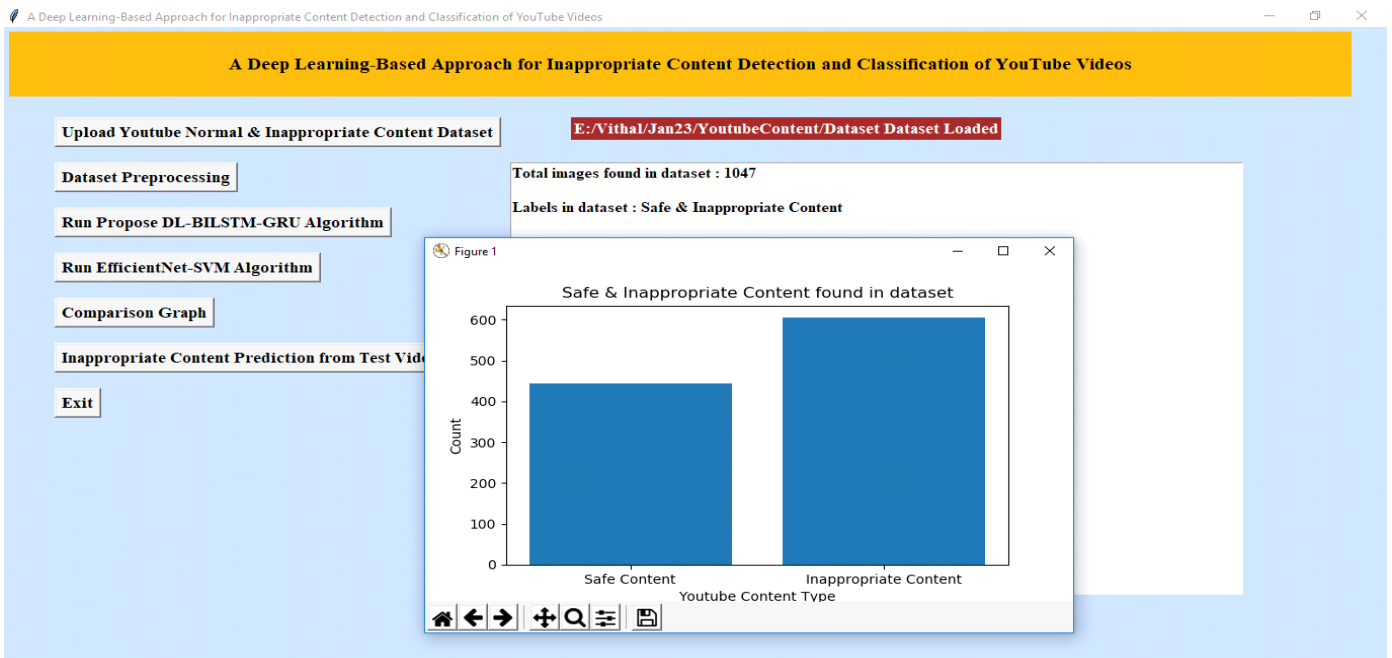


Fig : 6.6 - Dataset: 1047 images. Graph: X-axis (Safe & Inappropriate), Y-axis (count). Processing complete. Click ‘Run Proposed DL-BILSTM-GRU Algorithm’ to train and view output.

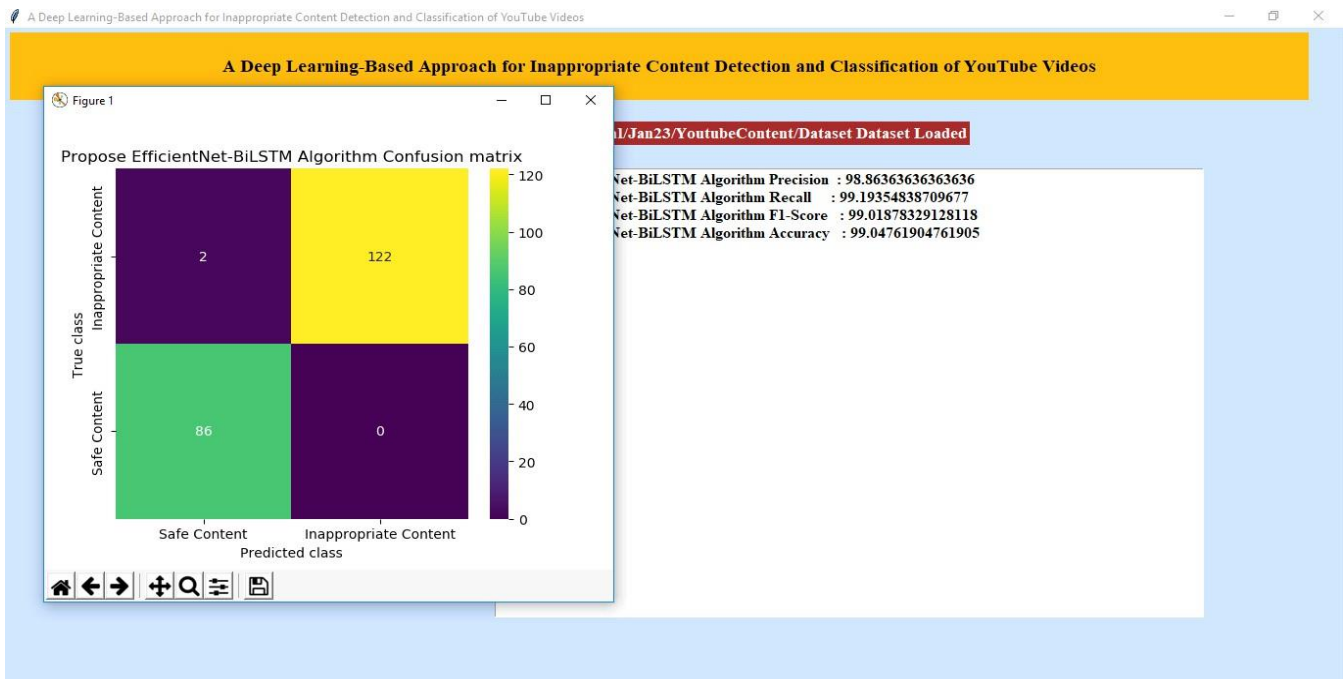


Fig : 6.7 - In above screen with propose EfficientNetB7-BI-LSTM we got 99.04% accuracy and in confusion matrix graph x-axis represents Predicted Labels and y-axis represents True Labels and green and yellow boxes contains correct prediction count and blue boxes contains incorrect prediction count which is 2 only. Now close above graph and then click on 'Run EfficientNet-SVM Algorithm' button to get below output

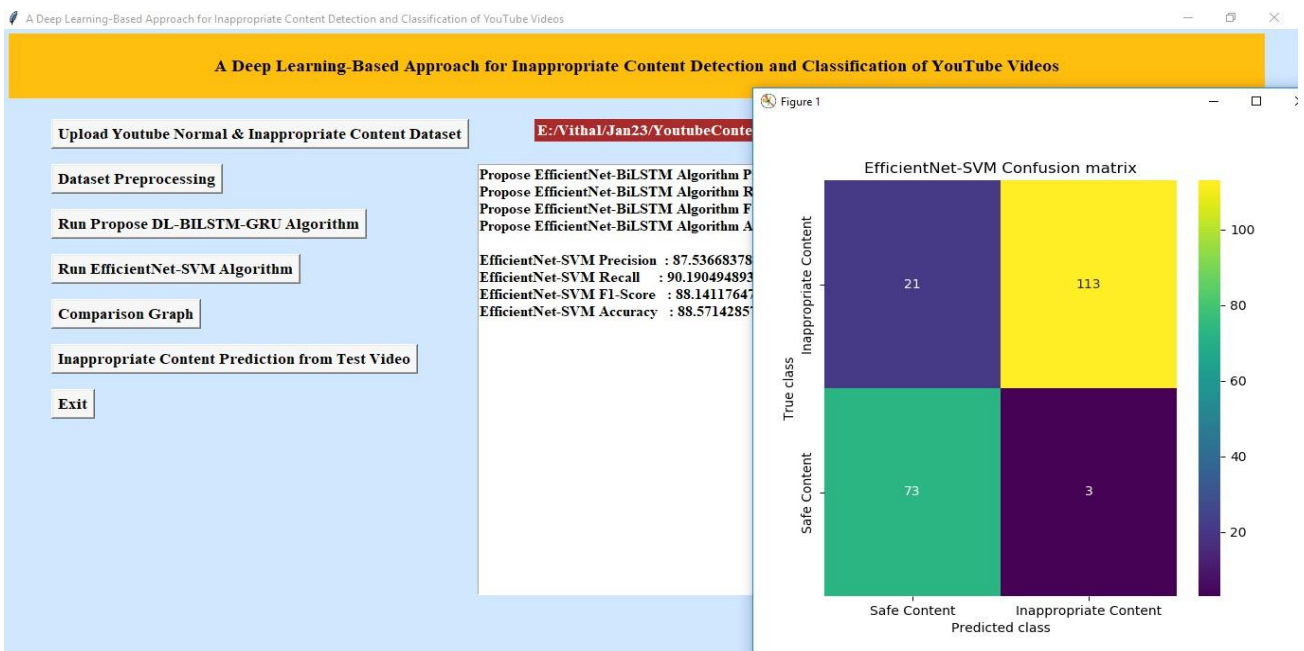


Fig : 6.8 - In above screen with EfficientNetB7-SVM we got 88% accuracy and in confusion matrix graph we can see in blue boxes that SVM predicted total 24 incorrect prediction so its accuracy is less. Now close above graph and then click on ‘Comparison Graph’ button to get below output

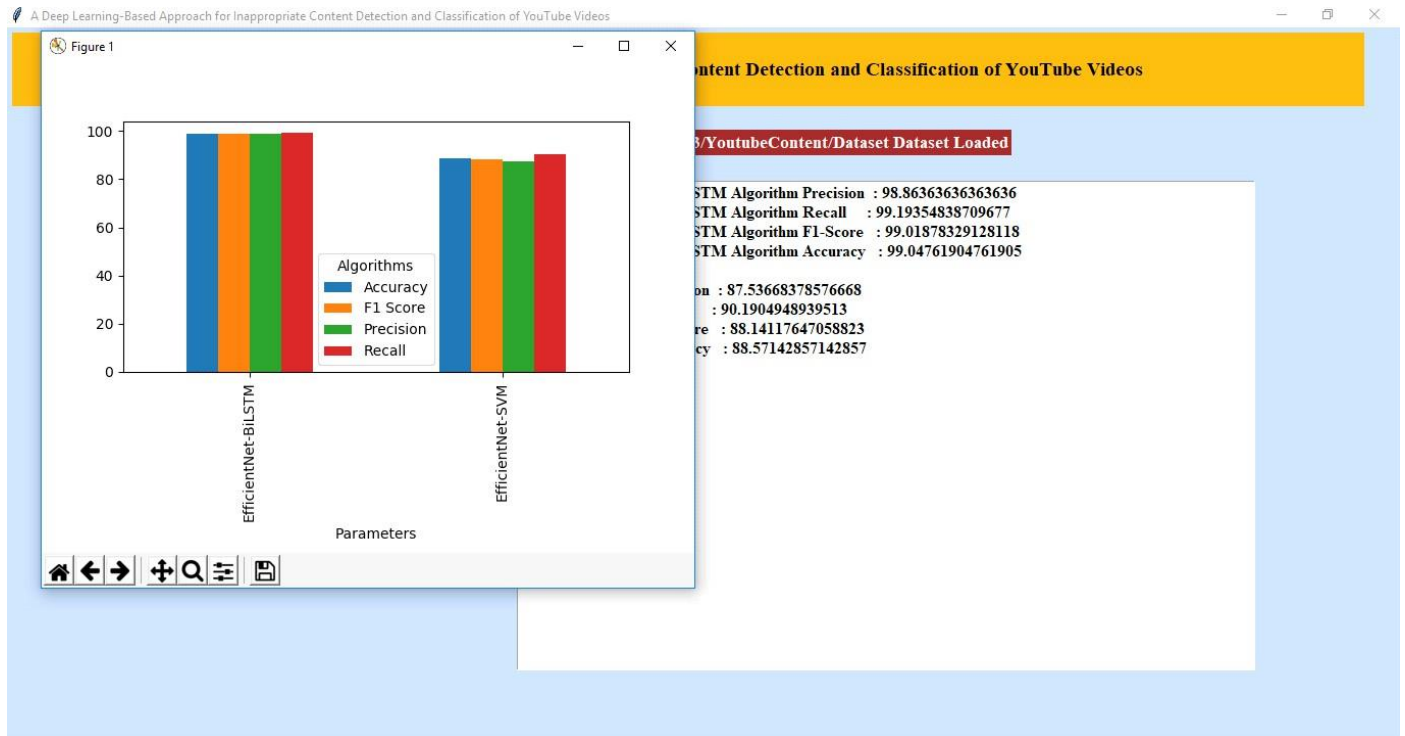


Fig : 6.9 - In above graph x-axis represents algorithm names and y-axis represents accuracy and other metrics in different colour bars. In both algorithms propose EfficientNetB7-BI-LSTM got high accuracy. Now close above graph and then click on ‘Inappropriate Content Prediction from Test Video’ button to upload test video and classify it as Safe or inappropriate.

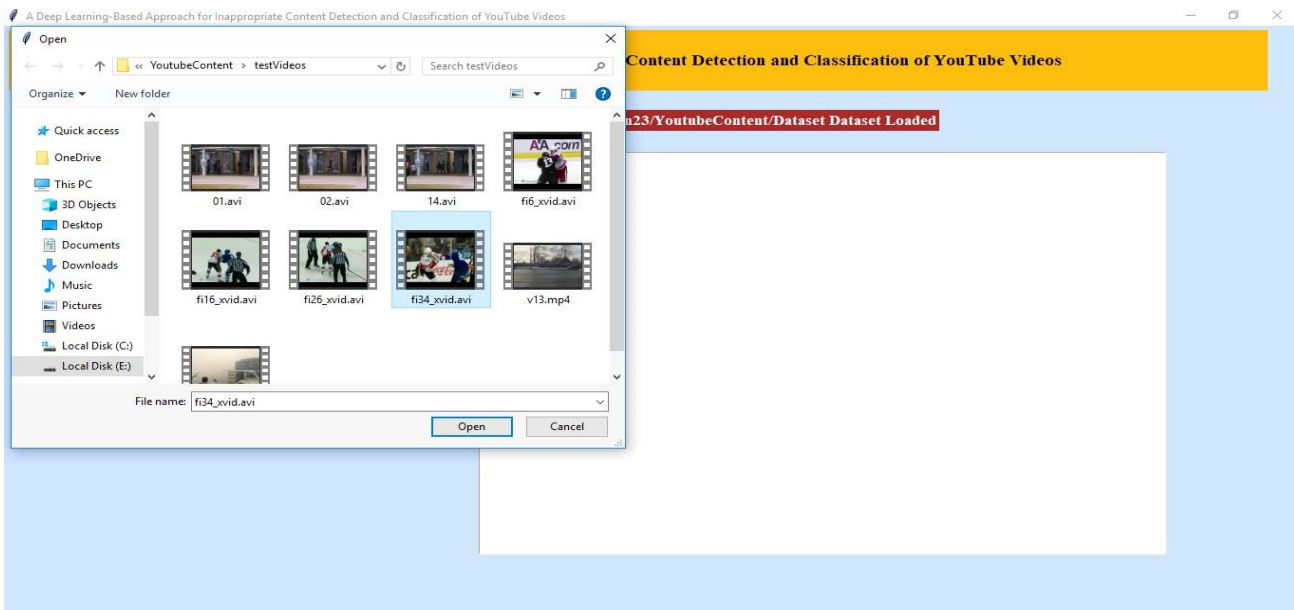


Fig : 6.10 - In above screen selecting and uploading video and then click on “Open’ button to play video and perform classification

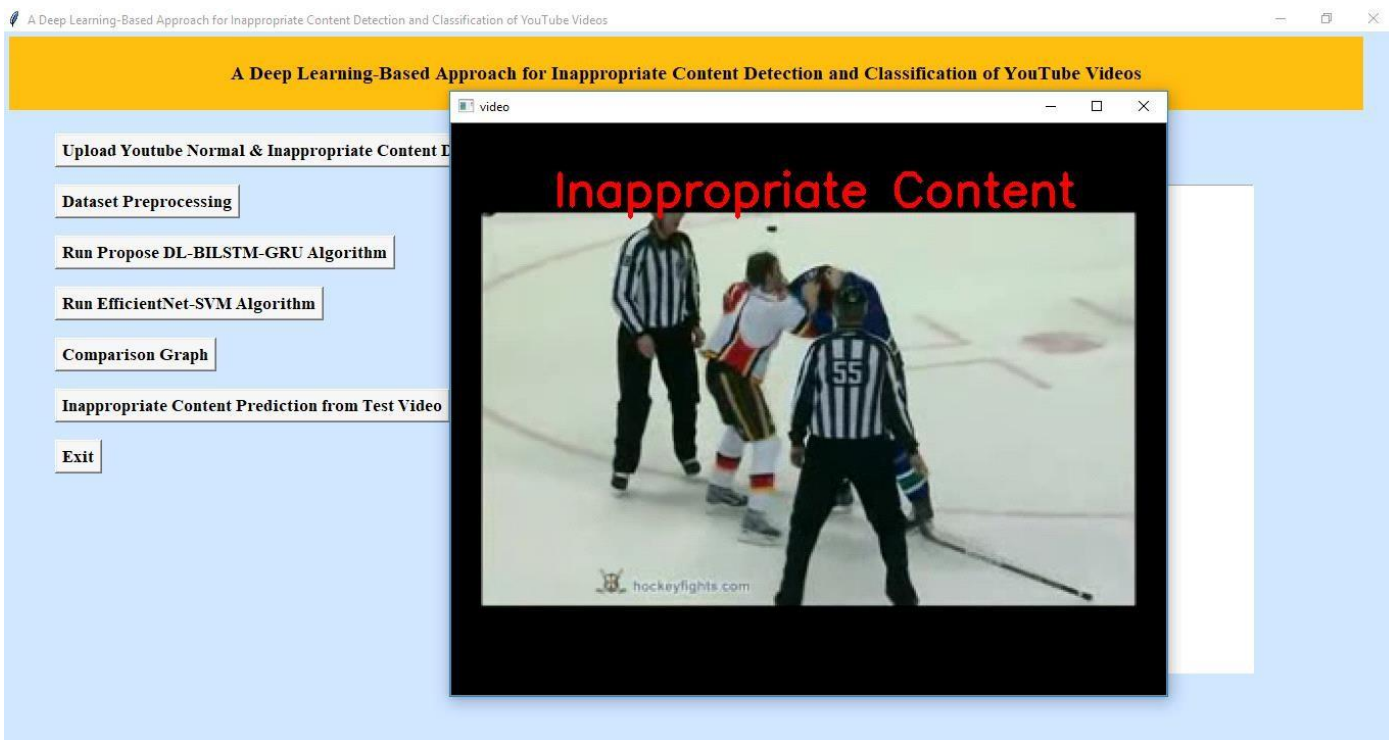


Fig : 6.11 - In above screen propose algorithm evaluating playing video and then detecting and classifying it as ‘Inappropriate Content’

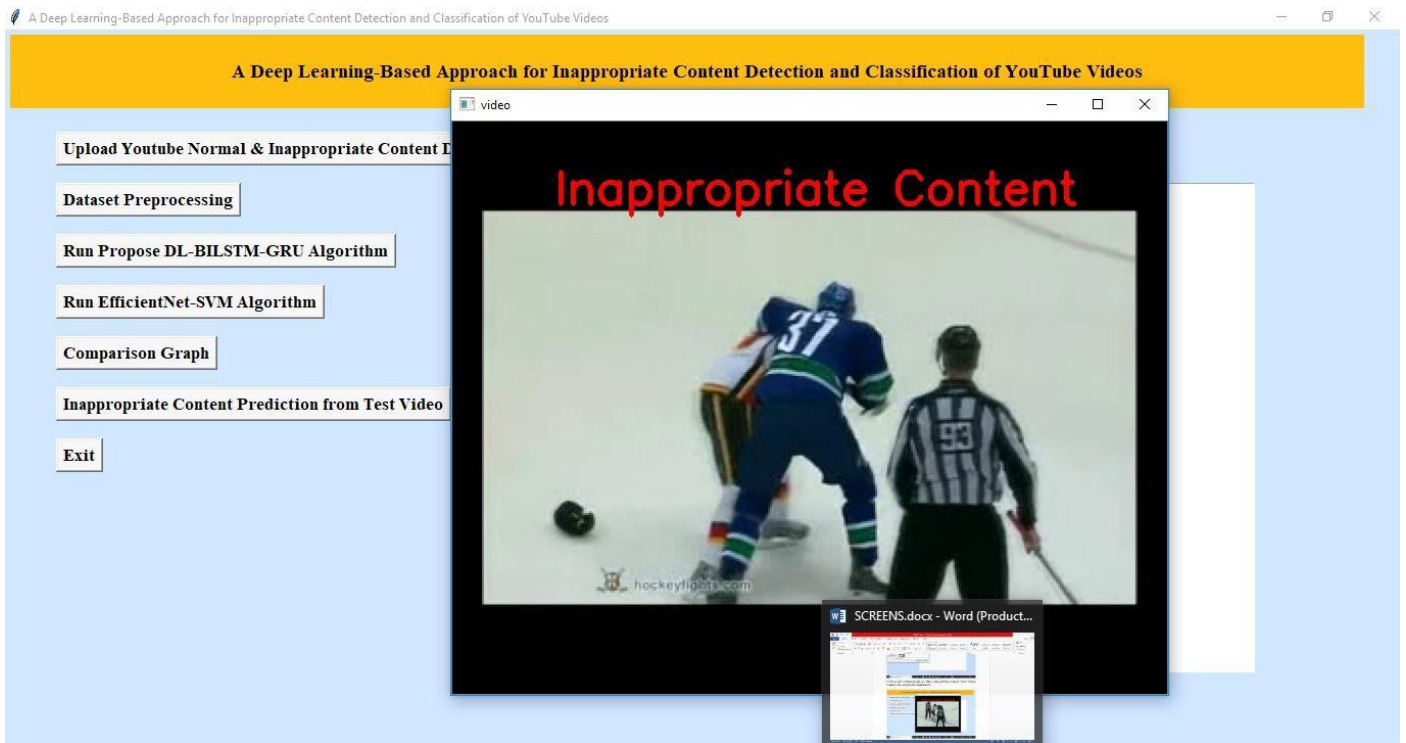


Fig : 6.12 - In above video also we got classification output

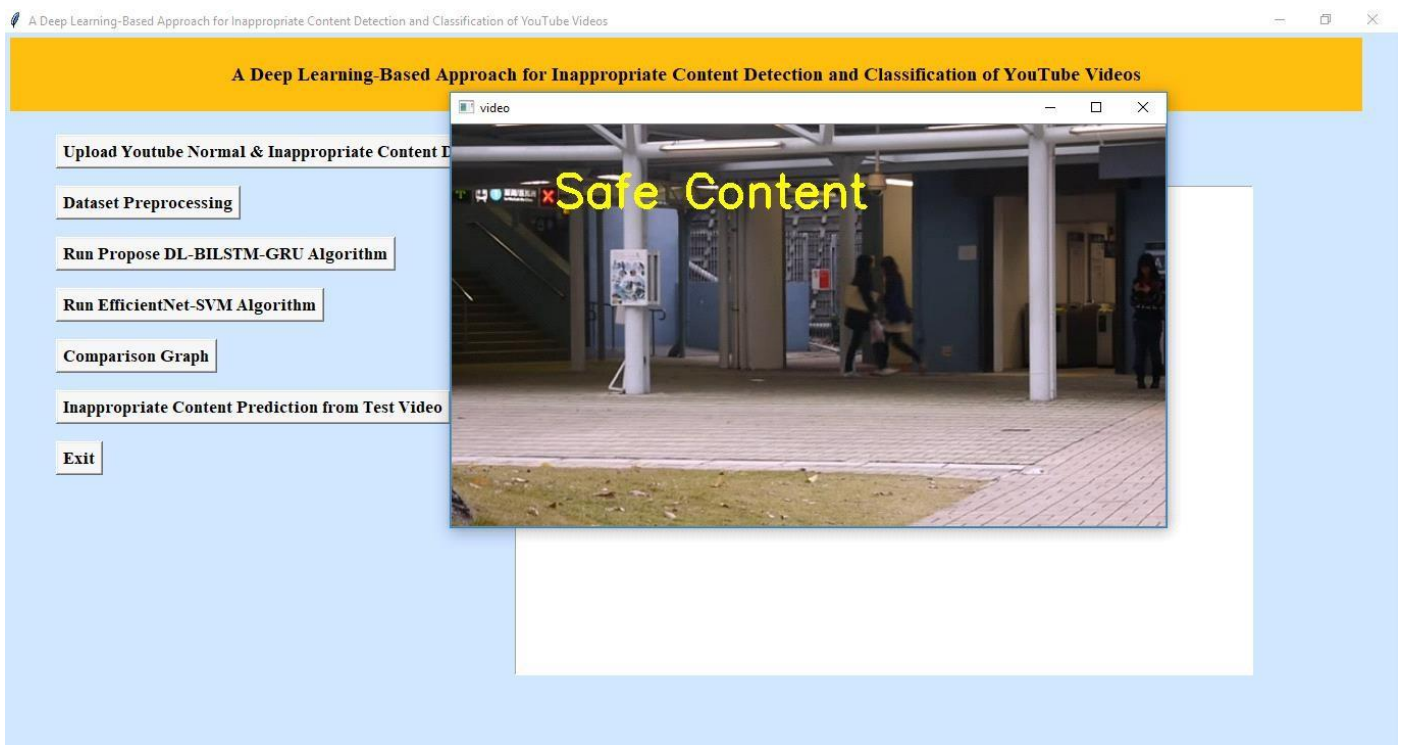


Fig : 6.13 - In above we got result as Safe Content

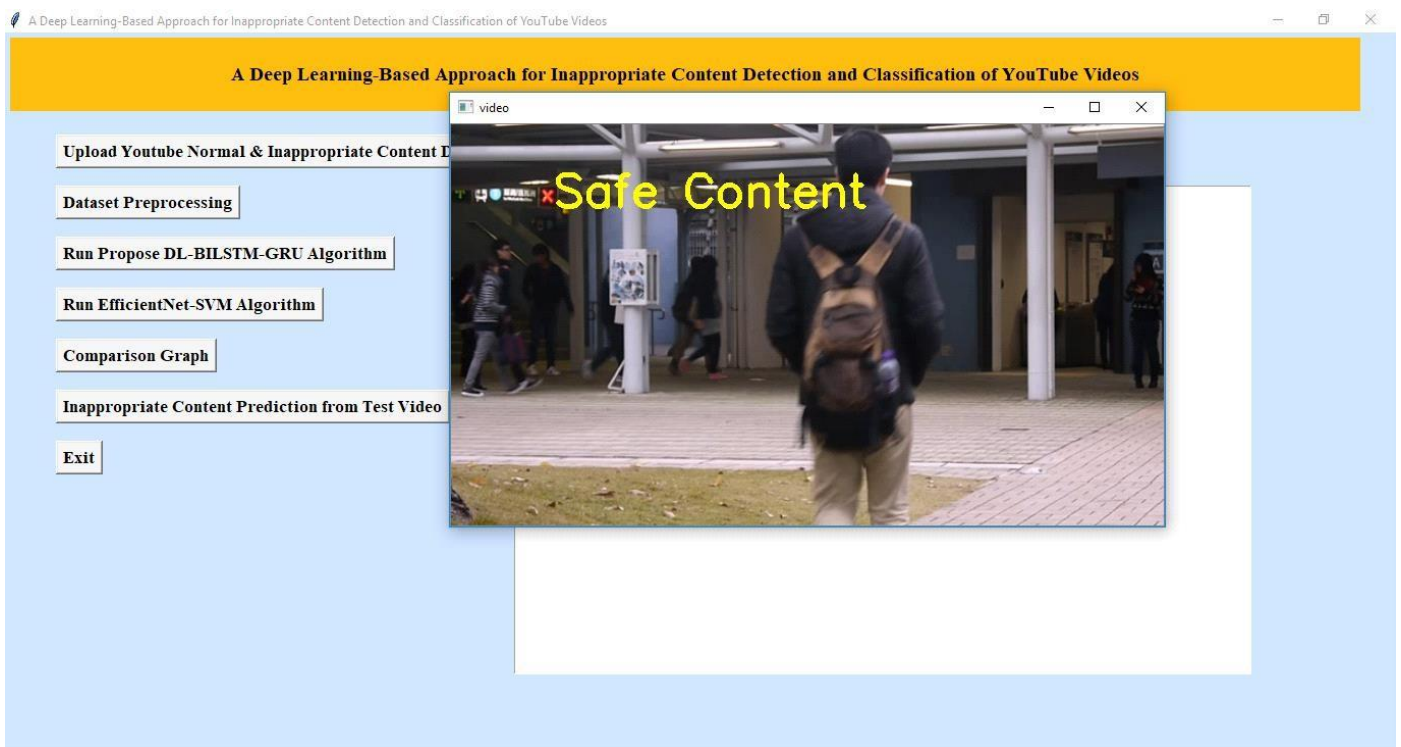


Fig : 6.14 - In above screen we got output as Safe Content as peoples are only moving in the video. Similarly you can upload and test other videos also

CHAPTER 7: ADVANTAGES AND APPLICATIONS

7.1: Advantages:

- **Automated Moderation** – Reduces manual effort in content review.
- **Real-Time Detection** – Quickly identifies inappropriate content.
- **High Accuracy** – Deep learning improves classification precision.
- **Scalability** – Can handle large volumes of YouTube videos.
- **Multi-Category Classification** – Differentiates various types of inappropriate content.
- **Adaptability** – Can be fine-tuned for evolving content trends.
- **User Protection** – Enhances online safety by filtering harmful content.
- **Compliance Support** – Helps platforms adhere to content regulations

7.2: Applications

- **YouTube Content Moderation** – Automates policy enforcement on YouTube.
- **Social Media Platforms** – Filters inappropriate videos on various platforms.
- **Parental Control Systems** – Blocks harmful content for younger audiences.
- **Corporate Training Filters** – Ensures workplace-friendly video content.
- **Educational Platforms** – Prevents exposure to inappropriate materials.
- **Video Streaming Services** – Maintains content integrity on OTT platforms.
- **Government Monitoring** – Assists in detecting extremist or illegal content.
- **Online Advertising** – Prevents ads from appearing on unsuitable videos.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION:

In this paper, a novel deep learning-based framework is proposed for child inappropriate video content detection and classification. Transfer learning using EfficientNet-B7 architecture is employed to extract the features of videos. The extracted video features are processed through the BiLSTM network, where the model learns the effective video representations and performs multiclass video classification.

All evaluation experiments are performed by using a manually annotated cartoon video dataset of 111,156 video clips collected from YouTube. The evaluation results indicated that proposed framework of EfficientNet-BiLSTM (with hidden units = 128) exhibits higher performance (accuracy = 95.66%) than other experimented models including EfficientNet-FC, EfficientNet-SVM, EfficientNet-KNN, EfficientNet-Random Forest, and EfficientNet-BiLSTM with attention mechanism-based models (with hidden units = 64, 128, 256, and 512).

Moreover, the performance comparison with existing state-of-the-art models also demonstrated that our BiLSTM-based framework surpassed other existing models and methods by achieving the highest recall score of 92.22%. The advantages of the proposed deep learning-based children inappropriate video content detection system are as follows:

Furthermore, our methodology to detect inappropriate children content from YouTube is independent of YouTube video metadata which can easily be altered by malicious uploaders to deceive the audiences. In the future, we intend to combine the temporal stream using optical flow frames with the spatial stream of the RGB frames to further improve the model performance by better understanding the global representations of videos. We also aim to increase the classification labels to target the different types of inappropriate children content of YouTube videos.

8.2 FUTURE SCOPE:

The future of inappropriate content detection and classification on YouTube using deep learning is promising, with several advancements on the horizon. The integration of multimodal analysis— combining video, audio, and textual data— can significantly enhance detection accuracy. Real-time adaptive learning will enable models to continuously update and adapt to emerging trends in inappropriate content. Expanding this system for cross-platform moderation will make it applicable to various social media and video-sharing platforms beyond YouTube. Additionally, incorporating user feedback mechanisms can improve model reliability by learning from real-time reports and corrections. With the increasing need for automated compliance monitoring, this technology can assist in enforcing global content policies and regulations. The implementation of explainable AI (XAI) will also help in making moderation decisions more transparent. Furthermore, advancements in edge AI can enable on- device filtering, reducing the dependency on cloud processing.

CHAPTER 9 : BIBLIOGRAPHY

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APPENDIX

Python

Python is a language that may be used for a wide range of tasks. From website designing to mathematics computation to workstation ui, it may be used in a variety of ways (Pygame, Panda3D). The pronunciation of the text is simple and or the spacing of the software is manageable. Reptile lets you can focuses on the subject at hand instead of just the vocabulary, which makes it a joy to programme in.

History of Python:

Guido Van Rossum, the creator of the Python programming language, has been working on it for a long time. For its original launch in February of that year, the design is made in the 1980s and 1990s.

Why Python was created?

He was part of the Protozoans team in the late 1980s, experimenting on distributed data processing systems. A system like ABC (which has a basic, good syntax) would be ideal since it would be able to acquire the Amoeba api calls. It was about this point he wanted to develop a language that could be extended. These led the creation of a new program which would have been subsequently

Why the name Python? termed Py.

No. It may not have been dubbed after a venomous snake for that matter. Rossum was a devotee of a late- '70s sitcom. In Monty Python's Fly Circus, the moniker "Py" was taken.

Features of Python:

A simple language which is easier to learn

Python's structure is incredibly clean and effective, making it ideal for beginners. Programming in Python is significantly simpler than it is in other languages such as C++, Java, or C#. Rather to becoming bogged down in the minutiae of grammar,

Python really encourages you to enjoy developing. As a beginner, Python is an ideal base.

Free and open-source

But for professional purposes, Py can always be freely used and distributed. In addition to being able to use and share Python-based applications, you may also make modifications to the Scripting language program code. Python is continually being improved by a massive group.

Portability

Py applications may be transferred of one machine after another and executed without modification. But most operating systems, like Dos, Mac OS X, and Pi, can execute it without issue.

Extensible and Embeddable

Assuming the service has to run quickly. Software program may be readily combined with code written in C/C++ as well as other language groups.

A high-level, interpreted language

Data structure, waste removal, and the like are not a concern, as they are in C/C++.

In the same way, Python basically turns your code into a syntax your device can comprehend when users run Python script. There's nothing to be concerned about in the background.

Large standard libraries to solve common tasks

As a result of Python's modules, programmers no longer should to develop all of the programs from scratch. As an illustration, consider the following: There's an issue with connecting a Web application to a Dbms. Using the MySQLdb plugin is as simple as typing `import MySQLdb` into your web browser. Dependency injection standard frameworks have been well vetted or are in widespread usage. As a result, you don't have to worry about it breaking your software.

Object-oriented

Objects are the foundation of the Python programming language. It's easier to address a complicated issue using object-oriented programming (OOP).

Object-oriented programming (OOP) allows you to break down large issues into less, more manageable pieces.

Applications of Python:

1. Simple Elegant Syntax

Programming language is enjoyable. Source code is easy to read as well as generate. Why? It's easy to read and understand. Let's have a look at this code as an illustration:

2. Not overly strict

Python does not need you to specify the form of a field. The hyphen is not required at the conclusion of such sentence.

Linux makes it mandatory that you adhere to best practises (like proper indentation). Newcomer will find it much simpler to study with the help of all these simple tips.

3. Expressiveness of the language

Python makes it possible to develop applications with more features for less code. There are too few than 500 keystrokes in this Myoclonic sport with software application and intelligent AI opponents. This is merely an illustration. Know the principles of Python and you'll be shocked at what else you can do.

4. Great Community and Support

Programming language has a strong number of people and developers that defend it. If you get stuck, there are just a lot of internet communities that may help .

LIBRARIES

- `pip install numpy==1.19.2`
- `pip install pandas==0.25.3`
- `pip install matplotlib==3.1.1`
- `pip install keras==2.3.1`
- `pip install tensorflow==1.14.0`
- `pip install h5py==2.10.0`
- `pip install protobuf==3.16.0`
- `pip install scikit-learn==0.22.2.post1`
- `pip install seaborn==0.10.1`
- `pip install opencv-python==4.1.1.26`
- `pip install opencv-contrib-python==4.3.0.36`
- `pip install keras-efficientnets==0.1.7`

Sample Code:

```
from tkinter import *s
import tkinter
from tkinter import filedialog
import matplotlib.pyplot as plt
from tkinter.filedialog import askopenfilename
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
import seaborn as sns
import pickle
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score
import os
from sklearn.metrics import accuracy_score
from sklearn import svm
import imutils
import numpy as np
import pandas as pd

import cv2
from keras.layers import Bidirectional, LSTM
from keras.utils.np_utils import to_categorical
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from keras.layers import MaxPooling2D
from keras.layers import Dense, Dropout, Activation, Flatten
from keras.layers import Convolution2D
from keras.models import Sequential
import pickle
from keras_efficientnets import EfficientNetB7
from keras.callbacks import ModelCheckpoint
```

```

from keras.models import Sequential, Model, load_model
from keras.layers import Conv2D, MaxPool2D, Flatten, Dense, InputLayer,
BatchNormalization, Dropout

main = tkinter.Tk()
main.title("A Deep Learning-Based Approach for Inappropriate Content Detection
and Classification of YouTube Videos")
main.geometry("1200x1200")
content_model = load_model("model/content_model.h5")

global X_train, X_test, y_train, y_test
global bilstm_model, cnn_model
global X, Y
accuracy = []
precision = []
recall = []
fscore = []

def uploadDataset():
    global filename, dataset, textdata, labels
    text.delete('1.0', END)
    filename = filedialog.askdirectory(initialdir=".")
    text.insert(END, str(filename) + " Dataset Loaded\n\n")
    pathlabel.config(text=str(filename) + " Dataset Loaded")

def preprocessDataset():
    text.delete('1.0', END)
    global X, Y
    if os.path.exists("model/X.txt.npy"):
        X = np.load('model/X.txt.npy')
        Y = np.load('model/Y.txt.npy')
    else:

```



```

X = []
Y = []
for root, dirs, directory in os.walk('Dataset'):
    for j in range(len(directory)):
        name = os.path.basename(root)
        if 'Thumbs.db' not in directory[j]:
            img = cv2.imread(root+"/"+directory[j])
            img = cv2.resize(img, (32, 32))
            label = 0
            if name == 'violence':
                label = 1
            X.append(img)
            Y.append(label)
            print(name+" "+directory[j]+" "+str(label))
X = np.asarray(X)
Y = np.asarray(Y)
np.save('model/X.txt',X)
np.save('model/Y.txt',Y)
X = X.astype('float32')
X = X/255
indices = np.arange(X.shape[0])
np.random.shuffle(indices)
X = X[indices]
Y = Y[indices]
unique, count = np.unique(Y, return_counts = True)
Y = to_categorical(Y)
text.insert(END,"Total images found in dataset : "+str(X.shape[0])+"\n\n")
text.insert(END,"Labels in dataset : Safe & Inappropriate Content")
text.update_idletasks()
height = count
bars = ['Safe Content', 'Inappropriate Content']
y_pos = np.arange(len(bars))
plt.bar(y_pos, height)
plt.xticks(y_pos, bars)
plt.title("Safe & Inappropriate Content found in dataset")

```

```

plt.xlabel("Youtube Content Type")
plt.ylabel("Count")
plt.show()

def calculateMetrics(algorithm, predict, target):
    acc = accuracy_score(target,predict)*100
    p = precision_score(target,predict,average='macro') * 100
    r = recall_score(target,predict,average='macro') * 100
    f = f1_score(target,predict,average='macro') * 100
    text.insert(END,algorithm+" Precision : "+str(p)+"\n")
    text.insert(END,algorithm+" Recall : "+str(r)+"\n")
    text.insert(END,algorithm+" F1-Score : "+str(f)+"\n")
    text.insert(END,algorithm+" Accuracy : "+str(acc)+"\n\n")
    text.update_idletasks()
    precision.append(p)
    accuracy.append(acc)
    recall.append(r)
    fscore.append(f)
    LABELS = ['Safe Content', 'Inappropriate Content']
    conf_matrix = confusion_matrix(target, predict)
    plt.figure(figsize =(6, 6))
    ax = sns.heatmap(conf_matrix, xticklabels = LABELS, yticklabels = LABELS,
annot = True, cmap="viridis" ,fmt ="g");
    ax.set_ylim([0,2])
    plt.title(algorithm+" Confusion matrix")
    plt.ylabel('True class')
    plt.xlabel('Predicted class')
    plt.show()

def runExistingSVM():
    global vectorizer, X, Y
    Y = np.argmax(Y, axis=1)
    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)
    global accuracy, precision, recall, fscore

```

```

svm_cls = svm.SVC(C=2.0, kernel="sigmoid")
svm_cls.fit(X_train,y_train)
predict = svm_cls.predict(X_test)
calculateMetrics("EfficientNet-SVM", predict, y_test)

def runProposeAlgorithms():
    text.delete('1.0', END)
    global bilstm_model, cnn_model, X, Y
    global accuracy, precision, recall, fscore
    accuracy.clear()
    precision.clear()
    recall.clear()
    fscore.clear()

    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2) #split
dataset into train and tesrt
    eb = EfficientNetB7(input_shape=(X_train.shape[1], X_train.shape[2],
X_train.shape[3]), include_top=False, weights=None)
    eb.trainable = False
    cnn_model = Sequential()
    cnn_model.add(eb)
    cnn_model.add(Convolution2D(32, (1, 1), input_shape = (X_train.shape[1],
X_train.shape[2], X_train.shape[3]), activation = 'relu'))
    cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
    cnn_model.add(Convolution2D(32, (1, 1), activation = 'relu'))
    cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
    cnn_model.add(Flatten())
    cnn_model.add(Dense(units = 256, activation = 'relu'))
    cnn_model.add(Dense(units = y_train.shape[1], activation = 'softmax'))
    cnn_model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics
= ['accuracy'])
    if os.path.exists("model/model_weights.hdf5") == False:
        model_check_point = ModelCheckpoint(filepath='model/model_weights.hdf5',
verbose = 1, save_best_only = True)
        hist = cnn_model.fit(X_train, y_train, batch_size = 32, epochs = 50,
validation_data=(X_test, y_test), callbacks=[model_check_point], verbose=1)

```

```

    f = open('model/history.pckl', 'wb')
    pickle.dump(hist.history, f)
    f.close()
else:
    cnn_model = load_model("model/model_weights.hdf5")
    cnn_model = Model(cnn_model.inputs, cnn_model.layers[-2].output)#creating
cnn model
    cnn_features = cnn_model.predict(X) #extracting cnn features from test data
    X = cnn_features
    print(X.shape)
    cnn_features = np.reshape(cnn_features, (cnn_features.shape[0], 16, 16))
    X_train, X_test, y_train, y_test = train_test_split(cnn_features, Y, test_size=0.2)
    bilstm_model = Sequential() #defining deep learning sequential object
    #adding LSTM bidirectional layer with 32 filters to filter given input X train data
to select relevant features
    bilstm_model.add(Bidirectional(LSTM(32, input_shape=(X_train.shape[1],
X_train.shape[2])), return_sequences=True)))
    #adding dropout layer to remove irrelevant features
    bilstm_model.add(Dropout(0.2))
    #adding another layer
    bilstm_model.add(Bidirectional(LSTM(32)))
    bilstm_model.add(Dropout(0.2))
    #defining output layer for prediction
    bilstm_model.add(Dense(y_train.shape[1], activation='softmax'))
    #compile GRU model
    bilstm_model.compile(loss='categorical_crossentropy', optimizer='adam',
metrics=['accuracy'])
    #start training model on train data and perform validation on test data
    if os.path.exists("model/bilstm_weights.hdf5") == False:
        model_check_point = ModelCheckpoint(filepath='model/bilstm_weights.hdf5',
verbose = 1, save_best_only = True)
        hist = bilstm_model.fit(X_train, y_train, batch_size = 16, epochs = 20,
validation_data=(X_test, y_test), callbacks=[model_check_point], verbose=1)
    else:
        bilstm_model = load_model("model/bilstm_weights.hdf5")
        predict = bilstm_model.predict(X_test)

```

```

predict = np.argmax(predict, axis=1)
target = np.argmax(y_test, axis=1)
calculateMetrics("Propose BiLSTM Algorithm", predict, target)

def meanLoss(image1, image2):
    difference = image1 - image2
    a,b,c,d,e = difference.shape
    n_samples = a*b*c*d*e
    sq_difference = difference**2
    Sum = sq_difference.sum()
    distance = np.sqrt(Sum)
    mean_distance = distance/n_samples
    return mean_distance

def predict():
    global content_model
    text.delete('1.0', END)
    filename = filedialog.askopenfilename(initialdir="testVideos")
    cap = cv2.VideoCapture(filename)
    print(cap.isOpened())
    while cap.isOpened():
        imagedump=[]
        ret,frame=cap.read()
        for i in range(10):
            ret,frame=cap.read()
            if frame is not None:
                image = imutils.resize(frame,width=700,height=600)
                frame=cv2.resize(frame, (227,227), interpolation = cv2.INTER_AREA)
                gray=0.2989*frame[:, :,0]+0.5870*frame[:, :,1]+0.1140*frame[:, :,2]
                gray=(gray-gray.mean())/gray.std()
                gray=np.clip(gray,0,1)
                imagedump.append(gray)
        imagedump=np.array(imagedump)
        imagedump.resize(227,227,10)
        imagedump=np.expand_dims(imagedump,axis=0)

```

```

imagedump=np.expand_dims(imagedump,axis=4)
output=content_model.predict(imagedump)
loss=meanLoss(imagedump,output)
if frame is not None:
    if frame.any() == None:
        print("none")
    else:
        break
if cv2.waitKey(10) & 0xFF==ord('q'):
    break
print(str(frame)+" "+str(loss))
if loss>0.00068:
    print('Inappropriate Content')
    cv2.putText(image,"Inappropriate
Content",(100,80),cv2.FONT_HERSHEY_SIMPLEX,1.5,(0,0,255),4)
else:
    cv2.putText(image,"Safe
Content",(100,80),cv2.FONT_HERSHEY_SIMPLEX,1.5,(0,255,255),4)
    cv2.imshow("video",image)
cap.release()
cv2.destroyAllWindows()
def graph():
    df =
pd.DataFrame([['BiLSTM','Precision',precision[0]],['BiLSTM','Recall',recall[0]],['BiLST
M','F1 Score',fscore[0]],['BiLSTM','Accuracy',accuracy[0]],
               ['EfficientNet-SVM','Precision',precision[1]],['EfficientNet-
SVM','Recall',recall[1]],['EfficientNet-SVM','F1 Score',fscore[1]],['EfficientNet-
SVM','Accuracy',accuracy[1]],
               ],columns=['Parameters','Algorithms','Value'])
    df.pivot("Parameters", "Algorithms", "Value").plot(kind='bar')
    plt.show()
def close():
    main.destroy()

font = ('times', 14, 'bold')

```

```
title = Label(main, text='A Deep Learning-Based Approach for Inappropriate  
Content Detection and Classification of YouTube Videos')
```

```
title.config(bg='DarkGoldenrod1', fg='black')
```

```
title.config(font=font)
```

```
title.config(height=3, width=120)
```

```
title.place(x=5,y=5)
```

```
font1 = ('times', 13, 'bold')
```

```
uploadButton = Button(main, text="Upload Youtube Normal & Inappropriate  
Content Dataset", command=uploadDataset)
```

```
uploadButton.place(x=50,y=100)
```

```
uploadButton.config(font=font1)
```

```
pathlabel = Label(main)
```

```
pathlabel.config(bg='brown', fg='white')
```

```
pathlabel.config(font=font1)
```

```
pathlabel.place(x=560,y=100)
```

```
preprocessButton = Button(main, text="Dataset Preprocessing",  
command=preprocessDataset)
```

```
preprocessButton.place(x=50,y=150)
```

```
preprocessButton.config(font=font1)
```

```
svmButton = Button(main, text="Run EfficientNet-SVM Algorithm",  
command=runExistingSVM)
```

```
svmButton.place(x=50,y=250)
```

```
svmButton.config(font=font1)
```

```
proposeButton = Button(main, text="Run Propose DL-BILSTM Algorithm",  
command=runProposeAlgorithms)
```

```
proposeButton.place(x=50,y=200)
```

```
proposeButton.config(font=font1)
```

```
graphButton = Button(main, text="Comparison Graph", command=graph)
```

```
graphButton.place(x=50,y=300)
```

```
graphButton.config(font=font1)
```

```
predictButton = Button(main, text="Inappropriate Content Prediction from Test  
Video", command=predict)  
predictButton.place(x=50,y=350)  
predictButton.config(font=font1)  
  
exitButton = Button(main, text="Exit", command=close)  
exitButton.place(x=50,y=400)  
exitButton.config(font=font1)  
  
font1 = ('times', 12, 'bold')  
text=Text(main,height=25,width=90)  
scroll=Scrollbar(text)  
text.configure(yscrollcommand=scroll.set)  
text.place(x=500,y=150)  
text.config(font=font1)  
  
main.config(bg='LightSteelBlue1')  
main.mainloop()
```


Institute Name:

Malla Reddy Institute of Technology & Science

Title of the Innovation/Prototype:

A Deep Learning-Based Approach for Inappropriate Content Detection and Classification of YouTube Videos

Team Lead Name:

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Team Lead Gender:

Male

FY of Development:

2024-25

Developed as part of:

Academic Requirement/Study Project

Innovation Type:

Product,Service

TRL LEVEL:

4

MRL Level:

MRL 2: Manufacturing concepts identified

IRL Level:

IRL 1: Basic Research (Need Identification & Peer Review Publications) & Completed First-Pass Business Model Canvas (BMC)

Theme:

ICT, cyber-physical systems, Blockchain, Cognitive computing, Cloud computing, AI & ML.,

Define the problem and its relevance to today's market / society / industry need:

The rapid growth of user-generated content on platforms like YouTube has led to a rise in inappropriate content, which can harm users, particularly vulnerable groups. Detecting and classifying such content is crucial for maintaining a safe and responsible online environment. A deep learning-based approach leverages advanced algorithms to automate this process, ensuring timely and accurate identification. This addresses the increasing need for effective content moderation in today's digital landscape, benefiting platforms, users, and regulatory bodies by enhancing user safety, promoting a healthier online space, and complying with industry regulations.

Describe the Solution / Proposed / Developed:

The proposed solution utilizes deep learning models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to detect and classify inappropriate content in YouTube videos. The system is trained on a large dataset of labeled video content, learning to identify harmful behaviors, explicit language, violence, or graphic content. By analyzing both video frames and audio tracks, the model ensures comprehensive content moderation. This automated approach enables real-time detection, reduces manual effort, and improves content safety, aligning with platform policies while fostering a secure environment for users.

Explain the uniqueness and distinctive features of the (product / process / service) solution:

The solution stands out through its integration of both visual and auditory analysis, using advanced deep learning models like CNNs and RNNs. This dual modality approach ensures more accurate and comprehensive detection of inappropriate content, as it processes both video frames and audio simultaneously. Additionally, the system adapts to evolving content trends through continuous learning, improving over time. Its real-time detection capability offers immediate content moderation, reducing manual oversight and enhancing scalability, making it particularly well-suited for large platforms such as YouTube, where vast amounts of content require rapid, reliable moderation.

How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any:

Unlike competitors' solutions that typically focus on either visual or auditory data alone, our approach combines both video frames and audio tracks for a more comprehensive analysis. This dual modality detection enhances accuracy and reduces false positives, ensuring more reliable content moderation. Additionally, our model continuously learns from evolving datasets, staying up-to-date with emerging trends in inappropriate content. Real-time detection and processing further differentiate our solution, offering immediate action without significant delays. This unique combination of depth, adaptability, and speed sets our solution apart in the competitive landscape of content moderation tools.

Is there any IP or Patentable Component associated with the Solution?:

No

Has the Solution Received any Innovation Grant/Seefund Support?:

No

Are there any Recognitions (National/International) Obtained by the Solution?:

No

***Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?:**

No

Had the Solution Received any Pre-Incubation/Incubation Support?:

No

Video URL:

<https://drive.google.com/file/d/17pUHT71Z2s9-RpCvuGM68S03bJ4g0mfA/view?usp=sharing>

Innovation Photograph:

[View File](#)